QUESTION ANALYSIS

EEE459

Compound Semiconductor and Hetero-Junction Devices

Question Analysis

Sessions 2021–22 to 2017–18 *4 papers, 82 questions, 148 formulas*

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1. At a glance:

This analysis covers **4 papers** (2021-22, 2020-21, 2018-19, 2017-18), with 82 questions and 510 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Bipolar transistors	108	21 %	16	HBT (24); Ebers-Moll (22); Injection efficiency (12)
Metal-semiconductor contacts	94	18 %	16	Schottky junction (53); Ohmic contacts (22); Barrier lowering (14)
Heterojunctions	81	16 %	13	Band diagrams (30); Fields and potentials (24); HEMT (21)
Alloys and strain	66	13 %	9	Lattice matching (26); Quaternary alloys (16); GaAsP band gaps (12)
JFETs	63	12 %	10	Pinch-off (31); Channel profiles (14); Transconductance (14)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2021-22	2020-21	2018-19	2017-18	Term tests	All
Band structure and materials	·	**17**	**6**	**7**	**12**	**42**
Density of states	·	**6**	·	·	**8**	14
Hole bands	·	**6**	·	·	**4**	10
Crystal structure	·	·	**6**	**3**	·	9
Materials	·	**5**	·	·	·	5
Effective mass	·	·	·	**4**	·	4
High-field transport	·	**11**	·	·	**18**	**29**
Mobility	·	**6**	·	·	**15**	21
Gunn effect	·	**5**	·	·	**3**	8
Alloys and strain	·	**20**	**8**	**27**	**11**	**66**
Lattice matching	·	**9**	·	**9**	**8**	26
Quaternary alloys	·	**7**	·	**9**	·	16
GaAsP band gaps	·	·	·	**9**	**3**	12
Epitaxy	·	·	**8**	·	·	8
Isoelectronic traps	·	**4**	·	·	·	4
PN junction	·	**5**	·	·	·	**5**
Graded junction	·	**5**	·	·	·	5
Metal-semiconductor contacts	·	**31**	**18**	**26**	**19**	**94**
Schottky junction	·	**13**	**6**	**19**	**15**	53
Ohmic contacts	·	**4**	**7**	**7**	**4**	22
Barrier lowering	·	**14**	·	·	·	14
Interface states	·	·	**5**	·	·	5
Heterojunctions	·	**20**	**34**	**17**	**10**	**81**
Band diagrams	·	**8**	**5**	**7**	**10**	30

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Topic and subtopic	2021-22	2020-21	2018-19	2017-18	Term tests	All
Fields and potentials	·	**12**	**12**	·	·	24
HEMT	·	·	**11**	**10**	·	21
Resonant tunnelling	·	·	**6**	·	·	6
MOS capacitor	·	·	**12**	·	**10**	**22**
MOS capacitor	·	·	**6**	·	**5**	11
Threshold voltage	·	·	**6**	·	**5**	11
Bipolar transistors	·	**17**	**51**	**40**	·	**108**
HBT	·	·	**13**	**11**	·	24
Ebers-Moll	·	**9**	**5**	**8**	·	22
Injection efficiency	·	·	**12**	·	·	12
Breakdown	·	·	·	**9**	·	9
Bandgap narrowing	·	·	·	**8**	·	8
Base minority carriers	·	·	**8**	·	·	8
High-level injection	·	**8**	·	·	·	8
Gummel-Poon	·	·	**7**	·	·	7
Non-ideal effects	·	·	**6**	·	·	6
TEBT	·	·	·	**4**	·	4
JFETs	·	**29**	**9**	**25**	·	**63**
Pinch-off	·	**15**	**5**	**11**	·	31
Channel profiles	·	**6**	·	**8**	·	14
Transconductance	·	**8**	·	**6**	·	14
Cut-off frequency	·	·	**4**	·	·	4

3. Priority list:

Must know: Metal-semiconductor contacts (94), Alloys and strain (66), Band structure and materials (42)

Likely: Bipolar transistors (108), Heterojunctions (81), JFETs (63), High-field transport (29)

Less often: MOS capacitor (22), PN junction (5)

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

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4. Topic pages:

4.1. Band structure and materials:

42 marks, 8 questions, in 4 of 4 papers. Book: Neamen Ch. 3; Sze Ch. 1.

Most often theory (4 of 8). Subtopics: Crystal structure; Effective mass; Hole bands; Materials; Density of states.

Formulas:

Subtopic	Formula	Equation	Used in
Crystal structure (9)	**Atoms per cell**	corner $\frac{1}{8}$, face $\frac{1}{2}$, edge $\frac{1}{4}$, inside 1	18-19 1(a)
	bcc packing	$4r = \sqrt{3}a$, fraction $\sqrt{3}\pi/8 = 0.68$	18-19 1(a)
	hcp ideal ratio	$c/a = \sqrt{8/3} = 1.633$	18-19 1(a)
	Zinc blende bond	$d = \sqrt{3}a/4$	18-19 1(a)
Effective mass (4)	**Effective mass**	$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$	17-18 1(b)
	Band edge parabola	$E - E_c = \frac{\hbar^2 k^2}{2m^*}$	17-18 1(b)
	Finite difference	$\frac{d^2E}{dk^2} \approx \frac{E_{i+1} - 2E_i + E_{i-1}}{(\Delta k)^2}$	17-18 1(b)
Hole bands (10)	**Parabolic band**	$E = \frac{\hbar^2 k^2}{2m^*}$	20-21 1(c), 20-21 7(c)
	Slope and curvature	$\frac{dE}{dk} = \frac{\hbar^2 k}{m^*}, \quad \frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*}$	20-21 1(c), 20-21 7(c)
	Si hole masses	$m_{hh,lh}[100] = \frac{m_0}{\gamma_1 \mp 2\gamma_2}, \quad m_{hh,lh}[111] = \frac{m_0}{\gamma_1 \mp 2\gamma_3}$	20-21 1(c), 20-21 7(c)
Materials (5)	**Emission wavelength**	$\lambda\ (\mu\text{m}) = \frac{1.24}{E_g\ (\text{eV})}$	20-21 2(a)
Density of states (14)	**Density of states, 3D**	$g_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$	20-21 3(b)
	Density of states, 2D	$g_{2D}(E) = \frac{m^*}{\pi\hbar^2}$ per subband	20-21 3(b)
	Density of states, 1D	$g_{1D}(E) \propto \frac{1}{\sqrt{E - E_n}}$	20-21 3(b)
	Parabolic band	$E = E_c + \frac{\hbar^2 k^2}{2m^*}$	21-22 7(b)
	Allowed k in a box	$k = \frac{2\pi n}{L}$	21-22 7(b)
	Infinite-well levels	$E_n = \frac{\hbar^2}{2m^*}\left(\frac{n\pi}{L_z}\right)^2$	21-22 7(b)
	3D (bulk)	$D_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$	21-22 7(b)

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Subtopic	Formula	Equation	Used in
	2D (well)	$D_{2D}(E) = \frac{m^*}{\pi\hbar^2}\sum_n H(E-E_n)$	21-22 7(b)
	1D (wire)	$D_{1D}(E) = \frac{1}{\pi\hbar}\sum_n \sqrt{\frac{2m^*}{E-E_n}}$	21-22 7(b)
	0D (dot)	$D_{0D}(E) = \sum_i 2\,\delta(E-E_i)$	21-22 7(b)
	General rule	$D \propto (E-E_n)^{d/2-1}$	21-22 7(b)

Symbols:

r = atom radius (Å) ↩

a = lattice constant (Å) ↩

c = height of the hcp cell (Å) ↩

d = nearest-neighbour (bond) distance (Å); number of free directions ↩

m^* = effective mass (kg) ↩

$\hbar$ = reduced Planck constant, $h/2\pi$ (J s) ↩

E = electron energy (J); energy from the band edge (J) ↩

k = wave number (m^{-1}) ↩

E_c = conduction-band edge (J) ↩

E_i = energy read from the E–k diagram at point i (J); discrete levels of the dot (J) ↩

Δk = step in k between the points (m^{-1}) ↩

m_{hh}, m_{lh} = heavy- and light-hole masses (kg)

m_0 = free-electron mass (kg) ↩

γ_1, γ_2, γ_3 = Luttinger parameters ↩

λ = wavelength of a photon of energy E_g (μm) ↩

E_g = band gap energy (eV) ↩

g_{3D} = states per unit energy per unit volume ($J^{-1}m^{-3}$) ↩

h = Planck's constant (J · s) ↩

g_{2D} = states per unit energy per unit area ($J^{-1}m^{-2}$) ↩

g_{1D} = states per unit energy per unit length ($J^{-1}m^{-1}$) ↩

E_n = edge of subband n (J); energy of level n above the band edge (J) ↩

n = integer ↩

L = box size (m) ↩

L_z = well width (m) ↩

D_{3D} = states per unit energy per unit volume ($J^{-1}m^{-3}$) ↩

D_{2D} = states per unit energy per unit area ($J^{-1}m^{-2}$) ↩

H = unit step function ↩

D_{1D} = states per unit energy per unit length ($J^{-1}m^{-1}$) ↩

D_{0D} = states per unit energy of one dot (J^{-1}) ↩

δ = Dirac delta function ↩

D = density of states ↩

Questions:

No.	Paper	Marks	Type	Question
Crystal structure				
1	18-19 1(a)	6	diagram	Draw the following crystal structures. (i) Hexagonal close packed crystal structure (ii) Zinc blende crystal structure. (iii) Body centered cubic crystal structure
2	17-18 1(a)	3	theory	Define mono crystalline, polycrystalline and amorphous materials.
Effective mass				
3	17-18 1(b)	4	derivation	What is effective mass? Explain a method with mathematical verification to determine effective mass from E-K diagram.
Hole bands				
4	20-21 1(c)	6	derivation	Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than for heavy holes.
5	20-21 TT1 7(c)	4	derivation repeats 20-21 1(c)	Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than that of heavy holes.
Materials				
6	20-21 2(a)	5	theory	Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it.

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No.	Paper	Marks	Type	Question
Density of states				
7	21-22 TT1 7(b)	8	theory repeats 20-21 3(b)	Explain how the density of states changes qualitatively as the dimensionality reduces from 3D $\rightarrow$ 2D $\rightarrow$ 1D $\rightarrow$ 0D. Sketch $D(E)$ vs E for each case.
8	20-21 3(b)	6	theory	Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \rightarrow 2D \rightarrow 1D$. Sketch $D(E)$ vs E for each case.

4.2. High-field transport:

29 marks, 5 questions, in 2 of 4 papers. Book: Sze Ch. 1, 10.

Most often theory (5 of 5). Subtopics: Mobility; Gunn effect.

Formulas:

Subtopic	Formula	Equation	Used in
Mobility (21)	**Effective mass**	$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$	17-18 7(b)
	Si conductivity mass	$m^*_{cn} = \frac{3}{1/m_l + 2/m_t} = 0.26\, m_0$	17-18 7(b)
	Mobility	$\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$	17-18 7(b)
	High-field mobility	$\mu(E) = v_d/E$; $v_{sat} \approx 10^7$ cm/s	17-18 7(b)
	Mobility	$\mu = \frac{e\,\tau}{m^*}$	20-21 1(a), 20-21 7(a)
Gunn effect (8)	Drift velocity, low field	$v_d = \mu E$	20-21 1(b), 20-21 7(b)
	Saturation velocity	$v_d \rightarrow v_{sat} \approx 10^7$ cm/s	20-21 1(b)
	Gunn oscillation	$f = \frac{v_d}{L}$	20-21 1(b), 20-21 7(b)

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No.	Paper	Marks	Type	Question

Symbols:

m^* = conductivity effective mass (kg) ↩

$\hbar$ = reduced Planck constant, $h/2\pi$ (J s) ↩

E = electron energy (J); electric field (V/cm) ↩

k = wave number (m^{-1})

m^*_{cn} = conductivity effective mass of electrons (kg) ↩

m_0 = free-electron mass (kg) ↩

m_l = longitudinal effective mass ($0.98\,m_0$) ↩

m_t = transverse effective mass ($0.19\,m_0$) ↩

μ = mobility ($cm^2/V\ s$) ↩

e = magnitude of the electron charge (C) ↩

τ = mean time between collisions (s) ↩

v_d = drift velocity (cm/s) ↩

v_{sat} = saturation velocity (cm/s) ↩

f = transit-time frequency (Hz) ↩

L = length of the GaAs sample (cm) ↩

Questions:

No.	Paper	Marks	Type	Question
Mobility				
1	20-21 1(a)	6	theory repeats 17-18 7(b)	Between Si and GaAs, which has higher electron mobility and why?
2	20-21 TT1 7(a)	3	theory repeats 20-21 1(a)	Why is the electron mobility of Si less than GaAs?
3	17-18 TT1 7(b)	12	theory	Why the effective mass of electrons in GaAs is different to that in Si? Which semiconductor has higher electron mobility? Discuss the variation of electron mobility in these semiconductors with electric field.
Gunn effect				
4	20-21 1(b)	5	theory	Discuss Velocity saturation and Gunn effect .
5	20-21 TT1 7(b)	3	theory repeats 20-21 1(b)	What is Gunn effect?

4.3. Alloys and strain:

66 marks, 9 questions, in 4 of 4 papers. Book: Neamen Ch. 14.

Most often theory (6 of 9). Subtopics: GaAsP band gaps; Quaternary alloys; Lattice matching; Epitaxy; Isoelectronic traps.

Formulas:

Subtopic	Formula	Equation	Used in
GaAsP band gaps (12)	**GaAsP gaps**	$E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$	17-18 1(c)
	Direct gap of GaAsP	$E_\Gamma(x) = 1.424 + 1.150x + 0.176x^2$	21-22 7(a)
	Indirect gap of GaAsP	$E_X(x) = 1.900 + 0.144x + 0.211x^2$	21-22 7(a)
	Crossover	$E_\Gamma(x_c) = E_X(x_c)$	21-22 7(a)
	Emission wavelength	$\lambda\ (\mu\text{m}) = \dfrac{1.24}{E_g\ (\text{eV})}$	21-22 7(a)
Quaternary alloys (16)	Photon energy	$E_g = \dfrac{hc}{\lambda} = \dfrac{1.24}{\lambda(\mu\text{m})}\ \text{eV}$	17-18 1(d)
	Vegard (quaternary)	$Q(x,y) = xyQ_1 + x(1-y)Q_2 + (1-x)yQ_3 + (1-x)(1-y)Q_4$	17-18 1(d)

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Subtopic	Formula	Equation	Used in
	Misfit	$f = \frac{a_L - a_s}{a_s}$	17-18 1(d), 17-18 3(b), 17-18 7(a)
	Critical thickness	$h_c \approx \frac{a_s}{2f}$	17-18 1(d), 17-18 3(b), 17-18 7(a)
	Vegard's law, ternary	$a(A_xB_{1-x}C) = x\,a_{AC} + (1-x)\,a_{BC}$	20-21 2(d)
	Quaternary $In_xGa_{1-x}As_ySb_{1-y}$	$P = xy\,P_{InAs} + x(1-y)\,P_{InSb} + (1-x)y\,P_{GaAs} + (1-x)(1-y)\,P_{GaSb}$	20-21 2(d)
Lattice matching (26)	**Vegard (ternary)**	$a(x) = x\,a_{AC} + (1-x)\,a_{BC}$	17-18 7(a)
	Matthews–Blakeslee	$h_c = \frac{b(1-\nu\cos^2\alpha)}{2\pi f(1+\nu)\cos\lambda}\left[\ln\frac{h_c}{b}+1\right]$	17-18 7(a)
	Lattice misfit	$f = \frac{a_{layer} - a_{sub}}{a_{sub}}$	20-21 2(b)
	Critical thickness, estimate	$h_c \sim \frac{a_{sub}}{2\lvert f\rvert}$	20-21 2(b)
Epitaxy (8)	Si CVD	$SiCl_4 + 2H_2 \rightarrow Si + 4HCl$	18-19 1(d)
	GaAs chloride VPE	$6GaCl + As_4 + 3H_2 \rightarrow 6GaAs + 6HCl$	18-19 1(d)

Symbols:

E_Γ = band gap to the Γ valley (eV) ↩

E_X = band gap to the X valleys (eV) ↩

x = P fraction in $GaAs_{1-x}P_x$; mole fraction of AC in $A_xB_{1-x}C$ ↩

x_c = P fraction where the two gaps are equal ↩

E_g = band gap energy (eV) ↩

λ = wavelength of a photon of energy E_g (μm); angle between the slip direction and the interface direction normal to the dislocation (60°) ↩

h = Planck's constant (J s)

c = speed of light (m/s)

Q = a property (lattice constant, Å) of $A_xB_{1-x}C_yD_{1-y}$ ↩

y = fraction of C on its sublattice; As fraction on the group V sublattice ↩

Q_1, Q_2, Q_3, Q_4 = the same property of AC, AD, BC and BD ↩

f = lattice misfit ↩

a_L = lattice constant of the layer (Å) ↩

a_s = lattice constant of the substrate (Å) ↩

h_c = critical thickness (Å) ↩

a = lattice constant of the alloy $A_xB_{1-x}C$ (Å) ↩

a_{AC}, a_{BC} = lattice constants of the binaries AC and BC (Å) ↩

P = lattice constant (Å) or band gap (eV) of the alloy ↩

P_{InAs}, P_{InSb}, P_{GaAs}, P_{GaSb} = the same property of the four binaries ↩

b = length of the Burgers vector, $a/\sqrt{2}$ (Å) ↩

ν = Poisson ratio ↩

α = angle between the dislocation line and its Burgers vector (60°) ↩

a_{layer} = lattice constant of the layer (Å) ↩

a_{sub} = lattice constant of the substrate (Å) ↩

Questions:

No.	Paper	Marks	Type	Question
GaAsP band gaps				
1	21-22 TT1 7(a)	3	theory repeats 17-18 1(c)	Show the compositional dependence of direct-energy gap Γ and indirect energy gap X for GaPAs. Which one is more stepper *[sic: should be steeper]* and why?
2	17-18 1(c)	9	theory	How the energy gap of $GaAs_{1-x}P_x$ is varied with mole fraction? Sketch the corresponding simplified E-k diagram. Discuss its light emission efficiency with and without iso-electric trap.

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No.	Paper	Marks	Type	Question
Quaternary alloys				
3	20-21 2(d)	7	numerical	Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å.
4	17-18 1(d)	9	numerical	For an experimental setup an epitaxial layer of $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. If a light of wavelength less than 1243 nm passes through the epitaxial layer then it is absorbed in the film, otherwise it transmits through the layer. The lattice parameter of the unstrained InGaAsSb is 5.93 Å. find x and y using the following table. Also determine the critical thickness for this system.
Lattice matching				
5	20-21 2(b)	9	theory	Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth.
6	17-18 3(b)	9	theory	Discuss the significance of critical thickness in epitaxial growth. Draw enegy *[sic: should be energy]* band structure of Si and GaAs and show optical transitions in such semiconductors.
7	17-18 TT1 7(a)	8	numerical	If $In_xGa_{1-x}As$ alloy is grown on GaAs substrate with critical thickness of 50Å, calculate the alloy composition. Lattice constants of InAs and GaAs is 6.058Å and 5.653Å respectively.
Epitaxy				
8	18-19 1(d)	8	theory	Explain the process of vapor phase epitaxy.
Isoelectronic traps				
9	20-21 2(c)	4	theory	Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$.

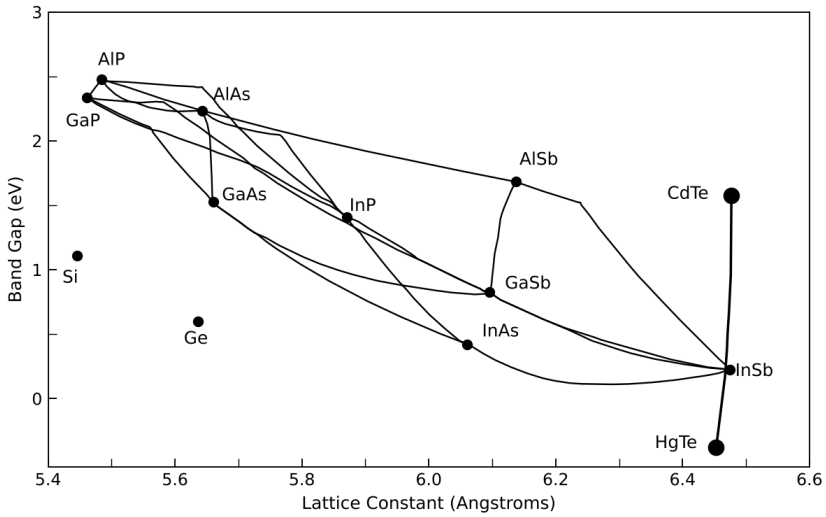


Figure 1: 20-21 2(d): Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)

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4.4. PN junction:

5 marks, 1 question, in 1 of 4 papers. Book: Neamen Ch. 7.

Most often theory (1 of 1). Subtopics: Graded junction.

Formulas:

Subtopic	Formula	Equation	Used in
Graded junction (5)	Net doping	$N_d - N_a = a\,x$	20-21 4(c)
	Depletion width	$W = \left[\frac{12\epsilon_s(V_{bi}+V_R)}{ea}\right]^{1/3}$	20-21 4(c)
	Junction capacitance	$C' = \left[\frac{ea\epsilon_s^2}{12(V_{bi}+V_R)}\right]^{1/3}$	20-21 4(c)

Symbols:

N_d, N_a = donor and acceptor concentrations (cm^{-3}) ↩

a = grading constant (cm^{-4}) ↩

x = distance from the metallurgical junction (cm) ↩

W = total depletion width (cm) ↩

ϵ_s = permittivity of the semiconductor (F/cm) ↩

V_{bi} = built-in potential (V) ↩

V_R = reverse bias (V) ↩

e = magnitude of the electron charge (C)

C' = capacitance per unit area (F/cm^2) ↩

Questions:

No.	Paper	Marks	Type	Question
Graded junction				
1	20-21 4(c)	5	theory	Describe how a linearly graded junction is formed.

4.5. Metal-semiconductor contacts:

94 marks, 16 questions, in 4 of 4 papers. Book: Neamen Ch. 9; Sze Ch. 3.

Most often theory (7 of 16). Subtopics: Schottky junction; Ohmic contacts; Interface states; Barrier lowering.

Formulas:

Subtopic	Formula	Equation	Used in
Schottky junction (53)	**Schottky barrier**	$\phi_{B0} = \phi_m - \chi$ (n); $\phi_{Bp} = \chi + E_g/e - \phi_m$ (p)	17-18 2(c), 17-18 3(a)
	Fermi level below E_c	$\phi_n = V_t \ln(N_c/N_d)$	17-18 2(c)
	Built-in potential	$V_{bi} = \phi_{B0} - \phi_n$	18-19 1(c), 18-19 7(b), 17-18 2(c)
	Depletion width	$x_n = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{eN_d}\right]^{1/2}$	20-21 3(c), 17-18 2(c), 17-18 3(c)
	Peak field	$\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s}$	18-19 1(c), 17-18 2(c)
	Barrier, n-type	$\phi_{B0} = \phi_m - \chi$	21-22 7(c), 20-21 3(c), 18-19 1(c), 18-19 3(d), 18-19 7(b)

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Subtopic	Formula	Equation	Used in
	Barrier, p-type	$\phi_{Bp} = \frac{E_g}{e} + \chi - \phi_m$	18-19 3(d)
	$E_c - E_F$ in n-type	$\phi_n = V_t \ln \frac{N_c}{N_d}$	18-19 1(c), 18-19 7(b)
	Depletion width	$x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$	21-22 7(c), 18-19 1(c)
	Built-in potential	$V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e} \ln \frac{N_c}{N_d}$	20-21 3(c)
	Maximum field	$E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{2(V_{bi} + V_R)}{x_n}$	20-21 3(c)
	Junction capacitance	$C' = \frac{\epsilon_s}{x_n} = \left[\frac{e\epsilon_s N_d}{2(V_{bi} + V_R)}\right]^{1/2}$	20-21 3(c)
	$E_c - E_F$ in n-type	$\phi_n = \frac{kT}{e} \ln \frac{N_c}{N_d}$	21-22 7(c)
	Built-in potential	$V_{bi} = \phi_{B0} - \phi_n = \phi_m - \phi_s$	21-22 7(c)
	Poisson's equation	$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s}$	21-22 7(c)
	Field, depletion region	$E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x)$	21-22 7(c)
	Metal sheet charge	$Q'_m = -eN_d x_n$	21-22 7(c)
	Maximum field	$\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s} = \left[\frac{2eN_d V_{bi}}{\epsilon_s}\right]^{1/2}$	21-22 7(c)
Ohmic contacts (22)	Tunnelling contact	$R_c \propto \exp\left[\frac{2\sqrt{\epsilon_s m^*}}{\hbar} \frac{\phi_{Bn}}{\sqrt{N_d}}\right]$	17-18 3(c)
	Specific contact resistance	$R_c = \left(\frac{\partial J}{\partial V}\right)^{-1}_{V=0}$	18-19 3(a)
	Thermionic R_c	$R_c = \frac{kT/e}{A^* T^2} \exp\left(\frac{e\phi_{Bn}}{kT}\right)$	18-19 3(a)
	Tunnelling R_c	$R_c \propto \exp\left(\frac{2\sqrt{\epsilon_s m_n^*}}{\hbar} \cdot \frac{\phi_{Bn}}{\sqrt{N_d}}\right)$	18-19 3(a), 18-19 7(a)
	Ohmic contact (ideal)	$\phi_m < \phi_s$ (n-type), $\phi_m > \phi_s$ (p-type)	20-21 5(d)
Interface states (5)	Interface states	$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right),\ \gamma = \frac{\epsilon_i}{\epsilon_i + e^2 \delta D_{it}}$	18-19 1(b)
Barrier lowering (14)	**Image-force lowering**	$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}$	20-21 3(d)
	Position of the peak	$x_m = \sqrt{\frac{e}{16\pi\epsilon_s E}}$	20-21 3(d)

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Subtopic	Formula	Equation	Used in
	Barrier with interface states	$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right)$	20-21 4(b)

Symbols:

ϕ_{B0} = barrier height for electrons, n-type (V); Schottky barrier height for electrons (V) ↩

ϕ_m = metal work function (V) ↩

χ = electron affinity of the semiconductor (V) ↩

ϕ_{Bp} = barrier height for holes, p-type (V); Schottky barrier height for holes (V) ↩

E_g = band gap energy (eV) ↩

e = magnitude of the electron charge (C) ↩

ϕ_s = work function of the semiconductor (V) ↩

V_t = thermal voltage, kT/e (V) ↩

N_c = effective density of states in the conduction band (cm^{-3}) ↩

N_d = donor concentration (cm^{-3}) ↩

V_{bi} = built-in potential, the band bending at zero bias (V) ↩

x_n = depletion width in the semiconductor (cm); depletion width at zero bias (cm) ↩

ϵ_s = permittivity of the semiconductor (F/cm) ↩

V_R = reverse bias (V) ↩

E_{max} = peak electric field, at the metal (V/cm); peak field, at the metal–semiconductor interface (V/cm); peak field, at the metal surface (V/cm) ↩

ϕ_n = depth of E_F below E_c (V) ↩

k = Boltzmann constant (J/K)

T = temperature (K) ↩

C' = capacitance per unit area (F/cm^2) ↩

E = electric field (V/cm); field at the interface (V/cm) ↩

x = distance from the metal (cm) ↩

ρ = charge density (C/cm^3) ↩

Q'_m = charge per unit area on the metal (C/cm^2) ↩

R_c = specific contact resistance (Ω cm^2) ↩

m^* = effective mass (kg) ↩

$\hbar$ = reduced Planck constant, $h/2\pi$ (J s) ↩

ϕ_{Bn} = barrier height for electrons (V) ↩

J = current density (A/cm^2) ↩

V = applied voltage (V) ↩

A^* = effective Richardson constant (A/cm^2K^2) ↩

m_n^* = electron effective mass (kg) ↩

γ = weight of the metal work function in the barrier; 1 without interface states, 0 when the Fermi level is pinned ↩

ϕ_0 = neutral level of the interface states, above E_v (V) ↩

ϵ_i = permittivity of the interfacial layer (F/cm) ↩

δ = thickness of the interfacial layer (cm) ↩

D_{it} = density of interface states ($cm^{-2}eV^{-1}$) ↩

$\Delta\phi$ = barrier lowering (V) ↩

x_m = distance of the barrier maximum from the metal (cm) ↩

Questions:

No.	Paper	Marks	Type	Question
Schottky junction				
1	21-22 TT1 7(c)	9	derivation	A metal-semiconductor (n-type) junction is formed where the work function of metal is greater than that of the semiconductor. With necessary assumptions, derive expression for electric field in both regions. Also sketch the electric field profile for the both regions.
2	20-21 3(a)	5	theory	Draw the ideal energy band diagram of a metal semiconductor junction under: (i) Reverse bias condition (ii) Forward bias condition
3	20-21 3(c)	8	numerical	Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ cm^{-3} at $T = 300$ K. Determine the (i) theoretical barrier height (ii) built-in potential barrier (iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and (iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ cm^{-3}.
4	18-19 1(c)	6	numerical	Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (metal work function 5.1 V) and n-type germanium (electron affinity 4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$

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No.	Paper	Marks	Type	Question
5	18-19 3(d)	0	numerical	Draw energy-band diagram of the following metal semiconductor junctions and mention whether it is Ohmic or Schottky then also find the ideal Schottky barrier height. (i) Al on top of n-GaAs (ii) Cu on top of n-GaAs (iii) Ag on top of p-GaAs (iv) Au on top of p-GaAs
6	18-19 TT1 7(b)	6	numerical repeats 18-19 1(c)	Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (5.1 V) and n-type germanium (4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$.
7	17-18 2(c)	10	numerical	Consider an ideal tungsten-to-n-type GaAs junction. Assume the GaAs is doped to a concentration of $N_d = 5 \times 10^{15}$ cm^{-3}. Determine the theoretical barrier height, the built-in potential barrier, and maximum electric field for the case of zero applied bias. (N_c = [unclear: 4.7×10^{17}] cm^{-3}, [unclear: ϵ_r] 13.1, $\phi_m = 4.55$, $\chi = 4.07$)
8	17-18 3(a)	9	diagram	Write the conditions necessary to form rectifying contacts in metal-semiconductor junctions. Draw the energy band diagrams of metal-semiconductor rectifying junctions under ideal conditions at thermal equilibrium. Draw diagrams for both n-type and p-type semiconductors.
Ohmic contacts				
9	20-21 5(d)	4	theory	Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact.
10	18-19 3(a)	4	theory	Define specific resistance. Explain which parameters affect specific resistance.
11	18-19 3(c)	3	theory repeats 18-19 7(a)	Briefly explain the types of ohmic contacts?
12	18-19 TT1 7(a)	4	theory	What is ohmic contact? Briefly explain the types of ohmic contacts.
13	17-18 3(c)	7	theory	Sketch the energy-band diagram of a tunneling junction. Why is this an ohmic contact?
Interface states				
14	18-19 1(b)	5	diagram	Sketch energy-band diagram of a metal-semiconductor junction with an interfacial layer and interface states.
Barrier lowering				
15	20-21 3(d)	6	numerical	Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm^{-1}. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height.
16	20-21 4(b)	8	theory	Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction: (i) Image-force-induced lowering (ii) Interface states

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4.6. Heterojunctions:

81 marks, 13 questions, in 3 of 4 papers. Book: Neamen Ch. 9.

Most often theory (4 of 13). Subtopics: Band diagrams; HEMT; Fields and potentials; Resonant tunnelling.

Formulas:

Subtopic	Formula	Equation	Used in
Band diagrams (30)	**Anderson offsets**	$\Delta E_c = e(\chi_n - \chi_P),\ \Delta E_v = \Delta E_g - \Delta E_c$	18-19 2(a), 18-19 2(d), 17-18 2(a), 17-18 2(b)
	Heterojunction V_{bi}	$eV_{bi} = \Delta E_v + kT \ln\left(\frac{p_{P0}N_{vn}}{p_{n0}N_{vP}}\right)$	17-18 2(a)
	Depletion widths	$x_n = \left[\frac{2\epsilon_n\epsilon_P N_{aP}V_{bi}}{eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2},\ N_{dn}x_n = N_{aP}x_P$	17-18 2(a)
	AlGaAs gap	$E_g(x) \approx 1.424 + (E_{g,\text{AlAs}} - 1.424)x$	17-18 2(b)
	AlGaAs band gap	$E_g(Al_xGa_{1-x}As) = 1.424 + 1.247x$ eV	20-21 1(d), 20-21 7(d)
	Band offsets (Anderson)	$\Delta E_c = e(\chi_n - \chi_w),\ \Delta E_g = \Delta E_c + \Delta E_v$	20-21 1(d), 20-21 7(d)
	Hole concentration	$p = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$	20-21 1(d), 20-21 7(d)
	Built-in potential, pN	$eV_{bi} = -\Delta E_v + kT \ln\left(\frac{p_P\, N_{vN}}{p_N\, N_{vP}}\right)$	20-21 1(d), 20-21 7(d)
HEMT (21)	Quantum-well levels	$E_n = \frac{\hbar^2\pi^2 n^2}{2m^* w^2}$	17-18 5(b)
	2DEG density	$n_s = \frac{\epsilon_N}{e(d+\Delta d)}(V_g - V_{off})$	18-19 6(b), 17-18 5(b), 17-18 5(d)
	HEMT off voltage	$V_{off} = \phi_B - \frac{\Delta E_c}{e} - V_{p2},\ V_{p2} = \frac{eN_d{d_d}^2}{2\epsilon_N}$	17-18 5(d)
	HEMT current	$I_D(\text{sat}) = \frac{\epsilon_N W}{d+\Delta d}(V_g - V_{off} - V_0)v_{sat}$	17-18 5(b)
	HEMT off voltage	$V_{off} = \phi_B - \frac{\Delta E_c}{e} - \frac{eN_d d_d^2}{2\epsilon_N}$	18-19 6(b)
Fields and potentials (24)	**Built-in potential (nP)**	$eV_{bi} = \Delta E_v + kT \ln\left(\frac{p_{P0}}{p_{n0}} \cdot \frac{N_{vn}}{N_{vP}}\right)$	18-19 2(d)
	Charge balance	$N_{dn}x_n = N_{aP}x_P$	18-19 2(b)
	Potential drops	$V_{bin} = \frac{eN_{dn}x_n^2}{2\epsilon_n},\ V_{biP} = \frac{eN_{aP}x_P^2}{2\epsilon_P}$	18-19 2(b)
	Depletion widths	$x_n = \left[\frac{2\epsilon_n\epsilon_P N_{aP}V_{bi}}{eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2},\ x_P = \left[\frac{2\epsilon_n\epsilon_P N_{dn}V_{bi}}{eN_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}$	18-19 2(b), 18-19 2(d)
	Total width	$W = \left[\frac{2\epsilon_n\epsilon_P(N_{dn}+N_{aP})^2 V_{bi}}{eN_{dn}N_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}$	18-19 2(b)

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Subtopic	Formula	Equation	Used in
	Field, N side	$E_1(x) = \frac{eN_{d1}}{\epsilon_1}(x + x_n)$	20-21 4(a)
	Field, P side	$E_2(x) = \frac{eN_{a2}}{\epsilon_2}(x_p - x)$	20-21 4(a)
	Interface condition	$\epsilon_1 E_1(0) = \epsilon_2 E_2(0) \Rightarrow N_{d1}x_n = N_{a2}x_p$	20-21 4(a)
	Potential split	$V_{bi1} = \frac{eN_{d1}x_n^2}{2\epsilon_1}$, $V_{bi2} = \frac{eN_{a2}x_p^2}{2\epsilon_2}$, $V_{bi} = V_{bi1} + V_{bi2}$	20-21 4(a)
	Depletion width, N side	$x_n = \left[\frac{2\epsilon_1\epsilon_2 N_{a2}V_{bi}}{eN_{d1}(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})}\right]^{1/2}$	20-21 4(a)
Resonant tunnelling (6)	RTD levels	$E_n \approx \frac{\hbar^2}{2m^*}\left(\frac{n\pi}{L_w}\right)^2$	18-19 3(b)

Symbols:

ΔE_c = conduction-band offset (eV)

e = magnitude of the electron charge (C) ↩

χ_n = electron affinity of the narrow-gap n side (V)

χ_P = electron affinity of the wide-gap P side (V) ↩

ΔE_v = valence-band offset (eV)

ΔE_g = difference of the two band gaps (eV) ↩

V_{bi} = built-in potential (V) ↩

k = Boltzmann constant (eV/K)

T = temperature (K)

p_{P0} = equilibrium hole concentration in the P side (cm^{-3})

p_{n0} = equilibrium hole concentration in the n side (cm^{-3})

N_{vn}, N_{vP} = effective densities of states in the valence band of the n and P sides (cm^{-3}) ↩

n_{in} = intrinsic carrier concentration of the n side (cm^{-3})

N_d = donor concentration of the n side (cm^{-3}); donor concentration in the AlGaAs (cm^{-3}) ↩

x_n, x_P = depletion widths in the n and P sides (cm) ↩

ϵ_n, ϵ_P = permittivities of the n and P sides (F/cm) ↩

N_{dn} = donor concentration in the n side (cm^{-3}) ↩

N_{aP} = acceptor concentration in the P side (cm^{-3}) ↩

E_g = band gap energy (eV) ↩

x = Al fraction in $Al_xGa_{1-x}As$; position, the interface at $x = 0$ (cm) ↩

$E_{g,\mathrm{AlAs}}$ = band gap of AlAs (eV) ↩

ΔE_c, ΔE_v = conduction- and valence-band offsets (eV) ↩

χ_n, χ_w = electron affinities of the narrow- and wide-gap materials (V) ↩

p = hole concentration (cm^{-3}) ↩

N_v = valence-band effective density of states (cm^{-3}) ↩

E_F = Fermi level (eV) ↩

E_v = valence-band edge (eV) ↩

p_P, p_N = hole concentrations in the P and N regions (cm^{-3}) ↩

N_{vP}, N_{vN} = valence-band densities of states of the P and N regions (cm^{-3}) ↩

E_n = energy of level n above the band edge in the well (J) ↩

$\hbar$ = reduced Planck constant (J s) ↩

n = level number (1, 2, ...) ↩

m^* = effective mass in the well (kg) ↩

w = well width (m) ↩

n_s = sheet density of the 2DEG (cm^{-2}) ↩

ϵ_N = permittivity of the AlGaAs (F/cm) ↩

d = AlGaAs thickness, $d_d + d_i$ (cm) ↩

d_d = doped AlGaAs thickness (cm) ↩

d_i = undoped spacer thickness (cm)

Δd = distance of the 2DEG from the interface (about 80 Å) ↩

V_g = gate voltage (V) ↩

V_{off} = off voltage (V) ↩

ϕ_B = Schottky barrier height of the gate (V) ↩

V_{p2} = pinch-off voltage of the doped AlGaAs (V) ↩

$I_D(\mathrm{sat})$ = saturated drain current (A) ↩

W = gate width (cm); total depletion width, $x_n + x_P$ (cm) ↩

V_0 = channel voltage at velocity saturation, LE_s (V) ↩

L = gate length (cm)

E_s = field at which the velocity saturates (V/cm)

v_{sat} = saturation velocity (cm/s) ↩

p_{P0}, p_{n0} = equilibrium hole concentrations in the P and n sides (cm^{-3}) ↩

n_i = intrinsic carrier concentration of the n side (cm^{-3})

$E_n(0)$, $E_P(0)$ = fields at the interface (V/cm)

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V_{bin}, V_{biP} = parts of V_{bi} across the n and P sides (V) ↩

N_{d1} = donor concentration of side 1 (cm^{-3}) ↩

N_{a2} = acceptor concentration of side 2 (cm^{-3}) ↩

V_R = reverse bias (V)

ϵ_1 = permittivity of side 1 (F/cm) ↩

ϵ_2 = permittivity of side 2 (F/cm) ↩

E_1 = electric field on the N side, side 1 (V/cm) ↩

E_2 = electric field on the P side, side 2 (V/cm) ↩

V_{bi1}, V_{bi2} = parts of V_{bi} dropped on each side (V) ↩

x_n = depletion width on the N side (cm) ↩

x_p = depletion width on the P side (cm) ↩

L_w = well width (m) ↩

Questions:

No.	Paper	Marks	Type	Question
Band diagrams				
1	20-21 1(d)	8	derivation	For a $GaAs - Al_xGa_{1-x}As$ heterojunction, determine the following: (i) Energy-Band diagram: Case-1: Before contact, Case-2: After contact (ii) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_P N_{VN}}{P_N N_{VP}}\right)$. Given, GaAs is doped P type, $Al_xGa_{1-x}As$ is doped N^+ type. Assume: $x = 0.25$ & $\Delta E_c = \frac{1}{2}\Delta E_g$ P_p = hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole respectively.
2	20-21 TT1 7(d)	10	derivation repeats 20-21 1(d)	For a GaAs-$Al_xGa_{(1-x)}As$ heterojunction, determine the following: (a) Energy-Band Diagram: (i) before contact (ii) After contact (b) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_p N_{VN}}{P_N N_{VP}}\right)$. Given, GaAs is doped p type, $Al_xGa_{(1-x)}As$ is doped N+ type. Assume: x=0.25 & $\Delta E_c = \frac{1}{2}\Delta E_g$. P_p = Hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole, respectively. $Hole\ concentration, P = N_V \exp\left(\frac{-(E_F - E_V)}{kT}\right)$
3	18-19 2(a)	5	diagram	Draw the band diagrams of a pN and a nN hetero-junction.
4	17-18 2(a)	7	numerical	What are the assumptions made in Anderson's model? Determine ΔE_C, ΔE_V and V_{bi} for an ideal n-Ge/p-GaAs heterojunction. Given that: Electron affinity of Ge = 4.13 eV, Electron affinity of GaAs = 4.07 eV, n_i = [unclear: 2.5×10^{13}] cm^{-3} (for Ge), $N_d = 1.2 \times 10^{16}$ cm^{-3}, and $N_a = 1 \times 10^{16}$ cm^{-3}.
5	17-18 2(b)	0	diagram	Sketch the energy-band diagram of an abrupt $Al_{0.3}Ga_{0.7}As/GaAs$ heterojunction for: (i) [unclear: N^+]-AlGaAs, intrinsic GaAs. (ii) P^+-AlGaAs, n-GaAs.
HEMT				
6	18-19 6(a)	4	theory	What are the advantages of High Electron Mobility Transistor (HEMT)?
7	18-19 6(b)	7	theory repeats 17-18 5(b)	With proper band diagram explain how High Electron Mobility Transistor (HEMT) operates.
8	17-18 5(b)	5	theory	What is 2DEG? With quantum well structure describe the operation of HEMT.
9	17-18 5(d)	5	numerical	Determine the two-dimensional electron concentration for an N AlGaAs – intrinsic GaAs heterojunction. Consider an $N - Al_{0.3}Ga_{0.7}As$ layer doped to 10^{18} cm^{-3} and having a thickness of 500 Å. Assume an undoped spacer layer of 20 Å. Let $\phi_B = 0.85$ V and $\Delta E_c/q = 0.22$ V. The relative dielectric constant of $Al_{0.3}Ga_{0.7}As$ is 12.2.
Fields and potentials				
10	20-21 4(a)	12	derivation	A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions.
11	18-19 2(b)	7	derivation	Derive the equation for the depletion region width of a hetero-junction.

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No.	Paper	Marks	Type	Question
12	18-19 2(d)	5	numerical	Determine the n-type doping concentration, and depletion region width in P-type side of semiconductor for an n-Ge and P-GaAs hetero-junction. Consider built in potential is 1.0 V and T = 300 K, n-type Ge electron affinity 4.13, N_{vn} = [unclear: 6 ×] 10^{18} cm^{-3}, $N_d = 10^{16}$ cm^{-3}, $n_i = 2.4 \times 10^{13}$ cm^{-3} and $\varepsilon_n = 16$. For P-type GaAs electron affinity 4.07, $N_{vP} = 7 \times 10^{18}$ cm^{-3}, $N_a = 10^{16}$ cm^{-3} and $\varepsilon_P = 13.1$
Resonant tunnelling				
13	18-19 3(b)	6	theory	With proper band diagram explain multiple negative differential resistance in a Resonant Tunneling Diode.

4.7. MOS capacitor:

22 marks, 4 questions, in 1 of 4 papers. Book: Neamen Ch. 10.

Most often diagram (2 of 4). Subtopics: MOS capacitor; Threshold voltage.

Formulas:

Subtopic	Formula	Equation	Used in
MOS capacitor (11)	Fermi potential	$\phi_{fp} = V_t \ln \frac{N_a}{n_i}$	18-19 4(c), 18-19 8(b)
	Maximum depletion width	$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$	18-19 4(c), 18-19 8(b)
	Oxide capacitance	$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$	18-19 4(c)
	Depletion C	$C' = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_d}$	18-19 4(c)
	Flat-band C	$C'_{FB} = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)L_D}$	18-19 4(c)
Threshold voltage (11)	**Flat-band voltage**	$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$	18-19 4(d)
	Threshold voltage	$V_{TN} = V_{FB} + \frac{\lvert Q'_{SD}(\max)\rvert}{C_{ox}} + 2\phi_{fp}$	18-19 4(d), 18-19 8(b)

Symbols:

V_t = thermal voltage, kT/e (V) ↩

N_a = substrate acceptor concentration (cm^{-3}) ↩

n_i = intrinsic carrier concentration (cm^{-3}) ↩

x_{dT} = maximum depletion width, at threshold (cm) ↩

ϵ_s = permittivity of the semiconductor (F/cm) ↩

e = magnitude of the electron charge (C)

ϕ_s = surface potential (V)

C_{ox} = oxide capacitance per unit area (F/cm^2) ↩

ϵ_{ox} = permittivity of the oxide, $3.9\epsilon_0$ (F/cm) ↩

t_{ox} = oxide thickness (cm) ↩

C' = capacitance per unit area in depletion (F/cm^2) ↩

x_d = depletion width (cm) ↩

C'_{FB} = flat-band capacitance per unit area (F/cm^2) ↩

L_D = extrinsic Debye length, $\sqrt{\epsilon_s V_t/(eN_a)}$ (cm) ↩

V_{FB} = flat-band voltage (V) ↩

ϕ_{ms} = metal–semiconductor work-function difference (V) ↩

Q'_{ss} = fixed oxide charge per unit area (C/cm^2) ↩

V_{TN} = threshold voltage of the n-channel device (V) ↩

$Q'_{SD}(\max)$ = maximum depletion charge per unit area (C/cm^2) ↩

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Questions:

No.	Paper	Marks	Type	Question
MOS capacitor				
1	18-19 4(c)	6	diagram	For an ideal Metal Oxide Semiconductor capacitor draw charge distribution, electric field distribution, potential distribution and capacitance versus voltage curve.
2	18-19 TT2 8(a)	5	diagram	Sketch the energy band diagram of the MOS capacitor under various bias conditions for both a p-type and s-type *[sic: should be n-type]* substrate.
Threshold voltage				
3	18-19 4(d)	6	derivation	Show the relation between Flat band voltage and threshold voltage of a MOS device.
4	18-19 TT2 8(b)	5	numerical	A p-type Si substrate at T = 300 K was doped at $N_a = 10^{14}\ cm^{-3}$ It has $Q'_{ss} = 10^{10}\ cm^{-2}$ and $t_{ox} = 30\ nm$ Determine the threshold voltage of an MOS system with a gold gate. Consider at T = 300 K, p-type Si electron affinity 4.01, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $N_c = 2.8 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3} and $\varepsilon_r = 11.7$ and gold p-Si work function $\varphi_{ms} = 0.07$ V.

4.8. Bipolar transistors:

108 marks, 16 questions, in 3 of 4 papers. Book: Neamen Ch. 12; Sze Ch. 5.

Most often theory (7 of 16). Subtopics: Ebers-Moll; Bandgap narrowing; Breakdown; TEBT; HBT; Base minority carriers; Injection efficiency; Non-ideal effects; Gummel-Poon; High-level injection.

Formulas:

Subtopic	Formula	Equation	Used in
Ebers-Moll (22)	**Ebers–Moll**	$I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R$	17-18 4(a)
	Reciprocity	$\alpha_F I_{ES} = \alpha_R I_{CS}$	20-21 6(a), 17-18 4(a)
	Ebers–Moll	$I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R,\ \alpha_F I_{ES} = \alpha_R I_{CS}$	18-19 6(c)
	Ebers–Moll currents	$I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R,\ I_B = I_E - I_C$	20-21 6(a)
	Diode currents	$I_F = I_{ES}\left[e^{V_{BE}/V_t} - 1\right],\ I_R = I_{CS}\left[e^{V_{BC}/V_t} - 1\right]$	20-21 6(a)
	Saturation voltage	$V_{CE}(sat) = V_t \ln\left[\frac{I_C(1-\alpha_R) + I_B}{\alpha_F I_B - (1-\alpha_F) I_C} \cdot \frac{\alpha_F}{\alpha_R}\right]$	20-21 6(a)
Bandgap narrowing (8)	Emitter minority holes	$p_{E0} = \frac{{n_i}^2}{N_E}$	17-18 4(d)
	Bandgap narrowing	$p'_{E0} = \frac{n_i^2}{N_E} \exp\left(\frac{\Delta E_g}{kT}\right)$	17-18 4(b), 17-18 4(d)
Breakdown (9)	Avalanche multiplication	$M = \frac{1}{1 - (V/BV_{CBO})^n}$	17-18 4(c)
	Open-base breakdown	$BV_{CEO} = BV_{CBO}(1-\alpha)^{1/n} = \frac{BV_{CBO}}{\sqrt[n]{\beta}}$	17-18 4(c)
TEBT (4)	Injection efficiency	$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}}$	17-18 5(a)
	Tunnelling probability	$T \approx \exp\left(-2d\sqrt{2m^* V_0}/\hbar\right)$	17-18 5(a)

Subtopic	Formula	Equation	Used in
HBT (24)	**Kirk effect**	$n_c = \frac{J_C}{ev_{sat}};\ J_K = ev_{sat}\left[N_C + \frac{2\epsilon_s(V_{CB}+V_{bi})}{eW_C{}^2}\right]$	17-18 5(c)
	HBT injection ratio	$\frac{J_{pE}}{J_{nE}} \propto \exp\left(-\frac{\Delta E_g}{kT}\right)$ (graded); $\exp(-\Delta E_v/kT)$ (abrupt)	17-18 5(c)
	HBT injection ratio	$\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nB}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$	18-19 4(b)
Base minority carriers (8)	**Base minority electrons**	$\delta n_B = \frac{n_{B0}(e^{eV_{BE}/kT}-1)\sinh\frac{x_B-x}{L_B} - n_{B0}\sinh\frac{x}{L_B}}{\sinh(x_B/L_B)}$	18-19 2(c)
Injection efficiency (12)	**Injection efficiency**	$\gamma = \left[1 + \frac{p_{E0}D_E L_B}{n_{B0}D_B L_E} \cdot \frac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}\right]^{-1}$	18-19 5(a), 18-19 5(d)
	Base transport factor	$\alpha_T = \frac{1}{\cosh(x_B/L_B)}$	18-19 5(d)
	Gain	$\alpha = \gamma\alpha_T\delta,\ \beta = \frac{\alpha}{1-\alpha}$	18-19 5(d)
Non-ideal effects (6)	Early effect	$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE}+V_A},\ r_o = \frac{V_A}{I_C}$	18-19 5(b)
Gummel-Poon (7)	Gummel–Poon current	$J_n = \frac{-eD_n n_i^2 e^{eV_{BE}/kT}}{\int_0^{x_B} p\,dx}$	18-19 5(c)
	Normalised base charge	$q_b = \frac{q_1}{2}(1+\sqrt{1+4q_2})$	18-19 5(c)

Symbols:

I_C, I_E = collector and emitter currents (A)

α_F = forward common-base current gain

α_R = reverse common-base current gain

I_F = forward diode current, $I_{ES}(e^{V_{BE}/V_t}-1)$ (A)

I_R = reverse diode current, $I_{CS}(e^{V_{BC}/V_t}-1)$ (A)

I_{ES}, I_{CS} = emitter and collector saturation currents (A); reverse saturation currents of the B–E and B–C junctions (A) ↩

V_{BE}, V_{BC} = base–emitter and base–collector voltages (V)

V_t = thermal voltage, kT/e (V) ↩

I_E = emitter current (A)

α_F, α_R = forward and reverse common-base current gains ↩

V_{BC} = base–collector voltage (V) ↩

I_C, I_E, I_B = collector, emitter and base currents (A) ↩

I_F, I_R = currents of the forward and reverse diodes (A) ↩

$V_{CE}(sat)$ = collector–emitter saturation voltage (V) ↩

p_{E0} = equilibrium hole concentration in the emitter (cm^{-3}) ↩

n_i = intrinsic carrier concentration (cm^{-3}) ↩

N_E = emitter doping (cm^{-3})

ΔE_g = band-gap narrowing (eV), from Fig. 1; band gap of the emitter minus that of the base (eV) ↩

k = Boltzmann constant (eV/K)

T = temperature (K); tunnelling probability ↩

M = avalanche multiplication factor ↩

V = reverse collector–base voltage (V) ↩

BV_{CBO} = collector–base breakdown voltage, emitter open (V) ↩

n = empirical exponent (3 to 6) ↩

BV_{CEO} = collector–emitter breakdown voltage, base open (V) ↩

α = common-base current gain ↩

β = common-emitter current gain, $\alpha/(1-\alpha)$ ↩

γ = emitter injection efficiency ↩

J_{nE} = electron current density injected into the base (A/cm^2) ↩

J_{pE} = hole current density injected into the emitter (A/cm^2) ↩

d = barrier thickness (m) ↩

m^* = effective mass (kg) ↩

V_0 = barrier height above the particle's energy (J) ↩

$\hbar$ = reduced Planck constant (J s) ↩

n_c = electron concentration in the collector depletion region (cm^{-3}) ↩

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J_C = collector current density (A/cm^2) ↩

e = magnitude of the electron charge (C) ↩

v_{sat} = saturation velocity (cm/s) ↩

J_K = current density at the onset of the Kirk effect (A/cm^2) ↩

N_C = collector doping (cm^{-3}) ↩

ϵ_s = permittivity of the semiconductor (F/cm) ↩

V_{CB} = collector–base reverse voltage (V) ↩

V_{bi} = built-in potential of the base–collector junction (V) ↩

W_C = thickness of the collector layer (cm) ↩

ΔE_v = valence-band offset (eV) ↩

N_{DE} = emitter donor concentration (cm^{-3}) ↩

N_{AB} = base acceptor concentration (cm^{-3}) ↩

W_E, W_B = emitter and base widths (cm) ↩

D_{nB} = electron diffusion coefficient in the base (cm^2/s) ↩

D_{pE} = hole diffusion coefficient in the emitter (cm^2/s) ↩

n_{iB}, n_{iE} = intrinsic concentrations of the base and emitter (cm^{-3})

δn_B = excess electron concentration in the base (cm^{-3}) ↩

n_{B0} = equilibrium electron concentration in the base (cm^{-3}) ↩

V_{BE} = base–emitter voltage (V) ↩

x = distance from the emitter edge of the base (cm) ↩

x_B = neutral base width (cm) ↩

L_B = electron diffusion length in the base, $\sqrt{D_B \tau_{B0}}$ (cm) ↩

D_B = electron diffusion coefficient in the base (cm^2/s) ↩

τ_{B0} = electron lifetime in the base (s)

D_E = hole diffusion coefficient in the emitter (cm^2/s) ↩

L_E = hole diffusion length in the emitter (cm) ↩

x_E = neutral emitter width (cm) ↩

N_B, N_E = base and emitter dopings (cm^{-3}) ↩

α_T = base transport factor ↩

δ = recombination factor ↩

I_C = collector current (A) ↩

V_{CE} = collector–emitter voltage (V) ↩

V_A = Early voltage (V) ↩

r_o = output resistance (Ω) ↩

J_n = electron current density through the base (A/cm^2) ↩

D_n = electron diffusion coefficient in the base (cm^2/s) ↩

p = hole concentration in the base (cm^{-3}) ↩

q_b = base charge over its zero-bias value ↩

q_1 = the Early-effect term ↩

q_2 = the high-injection term ↩

Questions:

No.	Paper	Marks	Type	Question
Ebers-Moll				
1	20-21 6(a)	9	derivation	Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model.
2	18-19 6(c)	5	diagram repeats 17-18 4(a)	Sketch the equivalent circuit for Ebers-Moll model as used in BJT.
3	17-18 4(a)	8	theory	From the Ebers-Moll equivalent circuit describe how a bipolar transistor operates.
Bandgap narrowing				
4	17-18 4(b)	3	theory	What is bandgap narrowing?
5	17-18 4(d)	5	numerical	Consider a silicon emitter at T = 300 K. Assume the emitter doping increases from 10^{18} to 10^{19} cm^{-3}. Determine the new value of p'_{E0} and determine the ratio p'_{E0}/p_{E0}.
Breakdown				
6	17-18 4(c)	9	derivation	Show that "The breakdown voltage in the open-base configuration is smaller, by the factor $\sqrt[n]{\beta}$ than the actual avalanche junction breakdown voltage."
TEBT				
7	17-18 5(a)	4	theory	With neat sketch, describe basic operation mechanism of TEBT.
HBT				
8	18-19 4(a)	6	diagram	Draw the typical structure of a Double Hetero-junction Bipolar Transistor Structure and their doping profile
9	18-19 4(b)	7	derivation	Describe the advantages of HBT over BJT. Show that for an HBT, the ratio of J_{nE} and J_{pE} can be expressed as $\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nE}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$ *[sic: should be D_{nB}].*

No.	Paper	Marks	Type	Question
10	17-18 5(c)	11	theory	With the help of energy-band diagram and electric field profile, explain Kirk effect in BJT. How Kirk effect is minimized in HBT? Compare abrupt and graded junction HBTs in terms of performance parameters.
Base minority carriers				
11	18-19 2(c)	8	derivation	For forward active mode derive the excess minority carrier concentration equation at base of a bipolar junction transistor
Injection efficiency				
12	18-19 5(a)	7	derivation	Derive the equation for emitter injection factor of a bipolar junction transistor.
13	18-19 5(d)	5	numerical	Consider a uniformly doped silicon npn bipolar transistor at T = 300 K with the following parameters: $D_B = 23$ cm^2/s, $D_E = 8$ cm^2/s, $\tau_{B0} = 2 \times 10^{-7}$ s, $\tau_{E0} = 8 \times 10^{-8}$ s, $N_B = 2 \times 10^{16}$ cm^{-3}, and $x_E = 0.35$ μm. The recombination factor has been determined to be $\delta = 0.9975$. The required common-emitter current gain is $\beta = 150$. A minimum neutral base width of $x_B = 0.80$ μm can be fabricated. (a) Determine an appropriate neutral base width and the minimum emitter doping concentration, N_E, to meet this specification. (b) Using the results of part (a), what are the values of α_T and γ?
Non-ideal effects				
14	18-19 5(b)	6	theory	Briefly explain the following non-ideal effects of a bipolar junction transistor. (i) Early effect (ii) Current Crowding
Gummel-Poon				
15	18-19 5(c)	7	theory	Describe the Gummel-Poon model of a bipolar junction transistor.
High-level injection				
16	20-21 6(b)	8	theory	Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved?

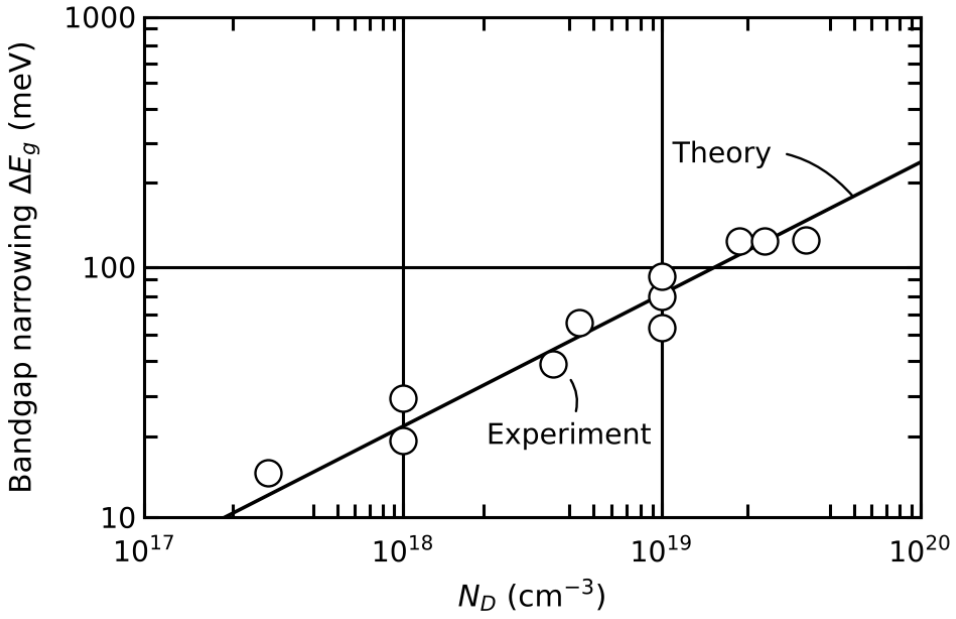


Figure 2: 17-18 4(d): Figure 1 of the paper (redrawn)

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4.9. JFETs:

63 marks, 10 questions, in 3 of 4 papers. Book: Neamen Ch. 13; Sze Ch. 7.

Most often numerical (4 of 10). Subtopics: Channel profiles; Pinch-off; Transconductance; Cut-off frequency.

Formulas:

Subtopic	Formula	Equation	Used in
Channel profiles (14)	Depletion thickness	$h = \left[\frac{2\epsilon_s(V_{bi} - V_{gs} + V(x))}{eN_d}\right]^{1/2}$	17-18 6(a), 17-18 6(b)
	Saturation drain voltage	$V_{d,sat} = V_g - V_p$	17-18 6(a)
	Saturation drain voltage	$V_{DS}(sat) = V_{p0} - (V_{bi} - V_{GS})$	20-21 5(a)
Pinch-off (31)	**Internal pinch-off**	$V_{p0} = \frac{ea^2N_d}{2\epsilon_s}$ (N_a for p channel)	17-18 6(b), 17-18 6(c)
	Pinch-off voltage	$V_p = V_{bi} - V_{p0}$ (n); $V_p = V_{p0} - V_{bi}$ (p)	17-18 6(c)
	pn built-in potential	$V_{bi} = V_t \ln\left(\frac{N_aN_d}{{n_i}^2}\right)$	17-18 6(c)
	MESFET threshold	$V_T = V_{bi} - V_{p0}$, $V_{bi} = \phi_{Bn} - \phi_n$	18-19 6(e)
	pn built-in potential	$V_{bi} = V_t \ln\frac{N_aN_d}{n_i^2}$	20-21 5(c)
	Space-charge width	$h = \left[\frac{2\epsilon_s(V_{bi} - V_{GS})}{eN_d}\right]^{1/2}$	20-21 5(b)
	Pinch-off voltage	$V_p = V_{bi} - V_{p0}$	20-21 5(b)
Transconductance (14)	Square law	$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_p}\right)^2$	17-18 6(d)
	Transconductance	$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right\|_{V_{DS}}$; $g_{ms} = -\frac{2I_{DSS}}{V_p}\left(1 - \frac{V_{GS}}{V_p}\right)$	17-18 6(d)
	Source resistance	$g'_m = \frac{g_m}{1 + g_m r_s}$	17-18 6(d)
	Pinch-off current	$I_{P1} = \frac{\mu_n(eN_d)^2Wa^3}{6\epsilon_sL}$	20-21 6(c)
	Saturation current	$I_{D1}(sat) = I_{P1}\left[1 - 3\left(\frac{V_{bi} - V_{GS}}{V_{p0}}\right)\left(1 - \frac{2}{3}\sqrt{\frac{V_{bi} - V_{GS}}{V_{p0}}}\right)\right]$	20-21 6(c)
	Transconductance	$g_{ms} = \frac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\frac{V_{bi} - V_{GS}}{V_{p0}}}\right)$	20-21 6(c)
Cut-off frequency (4)	**Cut-off frequency**	$f_T = \frac{g_m}{2\pi C_G}$; $f_T(\max) = \frac{e\mu_nN_da^2}{2\pi\epsilon_sL^2}$	18-19 6(d)
	Internal pinch-off	$V_{p0} = \frac{ea^2N_d}{2\epsilon_s}$	20-21 5(b), 18-19 6(d), 18-19 6(e)

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Symbols:

h = depletion thickness under the gate (cm); depletion width into the channel (cm) ↩

ϵ_s = permittivity of the semiconductor (F/cm) ↩

V_{bi} = built-in potential of the gate junction (V); built-in potential of the Schottky gate (V) ↩

V_{gs} = gate–source voltage (V) ↩

$V(x)$ = channel voltage at x (V) ↩

x = distance along the channel from the source (cm)

e = magnitude of the electron charge (C) ↩

N_d = donor concentration in the channel (cm^{-3}); channel donors (cm^{-3})

$V_{d,sat}$ = drain voltage at saturation (V) ↩

V_g = gate voltage (V) ↩

$V_{DS}(sat)$ = drain–source voltage at saturation (V) ↩

V_{p0} = internal pinch-off voltage (V) ↩

a = metallurgical channel thickness (cm)

N_a = acceptor concentration in a p channel (cm^{-3}); gate acceptors (cm^{-3})

V_p = pinch-off voltage: the gate voltage that just closes the channel at zero drain bias (V) ↩

V_t = thermal voltage, kT/e (V) ↩

N_a, N_d = acceptor and donor concentrations of the gate and channel (cm^{-3}) ↩

n_i = intrinsic carrier concentration (cm^{-3}) ↩

V_T = threshold voltage (V) ↩

ϕ_{Bn} = Schottky barrier height (V) ↩

V_{GS} = gate–source voltage (V) ↩

I_D = drain current (A) ↩

I_{DSS} = drain current at $V_{GS} = 0$ (A) ↩

g_m = transconductance (S) ↩

V_{DS} = drain–source voltage (V) ↩

g_{ms} = transconductance in saturation (S) ↩

g'_m = transconductance with a source resistance (S) ↩

r_s = source resistance (Ω) ↩

I_{P1} = pinch-off current (A) ↩

μ_n = electron mobility ($cm^2/V \cdot s$) ↩

W = channel width (cm)

L = channel length (cm); gate length (cm)

$I_{D1}(sat)$ = saturated drain current (A) ↩

f_T = cut-off frequency (Hz) ↩

C_G = gate capacitance (F) ↩

Questions:

No.	Paper	Marks	Type	Question
Channel profiles				
1	20-21 5(a)	6	theory repeats 17-18 6(a)	Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltage respectively) (i) $V_g > V_p$ and $V_d < V_{d,sat}$ (ii) $V_g > V_p$ and $V_d = V_{d,sat}$ (iii) $V_g > V_p$ and $V_d > V_{d,sat}$
2	17-18 6(a)	8	diagram	Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltages respectively) (i) $V_g > V_p$ and $V_d = 0V$ (ii) $V_g > V_p$ and $V_d < V_{d,sat}$ (iii) $V_g > V_p$ and $V_d \approx V_{d,sat}$ (iv) $V_g > V_p$ and $V_d > V_{d,sat}$
Pinch-off				
3	20-21 5(b)	8	derivation repeats 17-18 6(b)	Derive the formula of internal pinch-off voltage in an n-channel JFET.
4	20-21 5(c)	7	numerical	Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 3 \times 10^{15}$ cm^{-3}, $\epsilon_s = 13.1$, and $a = 0.70$ μm. Determine the forward-bias gate voltage required to open a channel region that is 0.10 μm thick with zero drain voltage.
5	18-19 6(e)	5	numerical	Determine the channel thickness of a GaAs MESFET to achieve threshold voltage $V_T = 0.1$ V. Consider an n-channel GaAs MESFET at T = 300 K with a gold Schottky barrier contact. Assume the barrier height is $\varphi_{Bn} = 0.89$ V. The n-channel doping is $N_d = 2 \times 10^{15}$ cm^{-3}.
6	17-18 6(b)	6	derivation	In an n-channel MESFET, (where the symbols have their usual meaning) show that, $V_{po} = \frac{ea^2 N_d}{2\epsilon_s}$
7	17-18 6(c)	5	numerical	The n^+p junction of a uniformly doped silicon p-channel JFET at T = 300 K has doping concentrations of $N_d = 10^{18}$ cm^{-3} and $N_a = 10^{16}$ cm^{-3}. The metallurgical channel thickness is $a = 0.40$ μm. Determine the internal pinchoff voltage and the pinchoff voltage of the JFET.

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No.	Paper	Marks	Type	Question
Transconductance				
8	20-21 6(c)	8	numerical	Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 1 \times 10^{16}$ cm^{-3}, $a = 0.75$ μm, $L = 10$ μm, $W = 30$ μm, and $\mu_n = 1000$ cm^2/(V s) (i) Calculate the maximum current in an n-channel JFET. (ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V.
9	17-18 6(d)	6	theory	Define transconductance for a JFET. Draw the small signal equivalent circuit of a JFET.
Cut-off frequency				
10	18-19 6(d)	4	derivation	Show that maximum cut off frequency of a MESFET is related to doping concentration.

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Band structure and materials				
1	21-22 TT1 7(b)	8	Density of states	$D(E)$ counts states per unit energy, from the allowed k points ($2\pi/L$ grid, 2 per point) and $E = \hbar^2k^2/2m^*$. 3D: shell $4\pi k^2 dk$, $D \propto \sqrt{E - E_c}$. 2D well: ring $2\pi k\,dk$, constant $m^*/\pi\hbar^2$ per subband, a staircase touching the 3D curve. 1D wire: two segments, $D \propto 1/\sqrt{E - E_n}$ spikes. 0D dot: delta functions. In general $D \propto (E - E_n)^{d/2-1}$.
2	20-21 2(a)	5	Materials	Near-IR (0.75 to 2 μm) uses III-V compounds: GaAs/AlGaAs (0.8 to 0.9 μm), InGaAsP and InGaAs on InP (1.3 and 1.55 μm, the fibre windows), Ge or InGaAs detectors. Reasons: band gaps 0.6 to 1.65 eV give $\lambda = 1.24/E_g$ in the near-IR; direct gaps give efficient emission; alloys tune the gap while lattice-matched to InP or GaAs.
3	20-21 3(b)	6	Density of states	$D(E)$ counts states per unit energy. 3D: $D \propto \sqrt{E - E_c}$. 2D quantum well: constant $m^*/\pi\hbar^2$ per subband, a staircase touching the 3D curve. 1D wire: $D \propto 1/\sqrt{E - E_n}$ spikes at each subband edge. 0D dot: delta functions. Confinement squeezes states into sharper features.
4	17-18 1(a)	3	Crystal structure	Single crystal: long-range order throughout (Si wafer). Polycrystalline: ordered grains of different orientation separated by grain boundaries (polysilicon). Amorphous: short-range order only (a-Si:H, SiO_2).
High-field transport				
5	20-21 1(a)	6	Mobility	GaAs: $\mu_n = 8500$ against 1350 cm^2/V s for Si. $\mu = e\tau/m^*$: GaAs has one sharply curved direct Γ valley with $m^* = 0.067\,m_0$; Si has six flat X valleys (conductivity mass $0.26\,m_0$) and intervalley scattering among them; GaAs also has weaker phonon scattering in the Γ valley.
6	20-21 1(b)	5	Gunn effect	Velocity saturation: hot carriers lose their extra energy to optical phonons, so v_d levels off near 10^7 cm/s (Si above about 3×10^4 V/cm). Gunn effect: in n-GaAs above about 3.2 kV/cm electrons transfer from the light Γ valley to the heavy L valley, v_d falls (negative differential mobility), and high-field domains form and travel, giving microwave current pulses.
7	20-21 TT1 7(a)	3	Mobility	Si 1350 against GaAs 8500 cm^2/V s. $\mu = e\tau/m^*$: Si electrons sit in six flat X valleys (conductivity mass $0.26\,m_0$), GaAs electrons in one sharp Γ valley ($0.067\,m_0$), 3.9 times lighter; Si also has intervalley scattering among its X valleys, so τ is shorter (0.20 against 0.32 ps).

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No.	Paper	Marks	Subtopic	Points to cover
8	20-21 TT1 7(b)	3	Gunn effect	In n-GaAs above about 3.2 kV/cm hot electrons transfer from the light Γ valley to the heavy L valleys 0.31 eV higher; v_d falls as E rises (negative differential mobility); high-field domains form at the cathode and drift to the anode, giving current pulses at $f = v_d/L$ (10 GHz for 10 μm).
9	17-18 TT1 7(b)	12	Mobility	$m^* = \hbar^2/(d^2E/dk^2)$: GaAs single sharp Γ valley, $0.067\,m_0$; Si six flat X ellipsoids, $m^*_{cn} = 0.26\,m_0$. GaAs mobility higher (8500 vs 1350 $\mathrm{cm^2/V \cdot s}$, $\mu = e\tau/m^*$). Field: constant at low field; Si saturates at 10^7 cm/s; GaAs peaks at 2×10^7 cm/s at 3.2 kV/cm, negative differential mobility (L valley), saturates.
Alloys and strain				
10	21-22 TT1 7(a)	3	GaAsP band gaps	Both gaps of $GaAs_{1-x}P_x$ rise with x: $E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$ eV; they cross at $x_c \approx 0.45$–0.48 (direct below, indirect above). The Γ gap is steeper (1.33 against 0.36 eV from GaAs to GaP): the s-like antibonding Γ state rises sharply as the smaller P shortens the bonds (large deformation potential); the p-like X state hardly moves.
11	20-21 2(b)	9	Lattice matching	Matching ($f = (a_L - a_S)/a_S \approx 0$) avoids misfit and threading dislocations, which are non-radiative recombination and scattering centres. A thin mismatched layer grows strained (pseudomorphic, tetragonal distortion) up to a critical thickness h_c (Matthews–Blakeslee), beyond which dislocations relax it; strain also shifts band edges and can be used on purpose.
12	20-21 2(c)	4	Isoelectronic traps	$GaAs_{1-x}P_x$ is direct for $x < 0.45$ and indirect above. Nitrogen on a P site is isoelectronic and forms a trap just below E_c; the electron is localised in space, so spread in k, and can recombine with a hole without a phonon; this makes indirect GaAsP:N (red-orange, yellow, green) LEDs efficient.
13	18-19 1(d)	8	Epitaxy	Single-crystal layer grown on a heated substrate from reacting gases (CVD). Si: $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$. GaAs chloride VPE: $AsCl_3 + H_2$ over a Ga boat (850 °C) gives GaCl and As_4, which deposit GaAs at 750 °C; MOVPE uses $Ga(CH_3)_3 + AsH_3$. Thickness by time and flow, doping by dopant gases, composition by mixing sources.
14	17-18 1(c)	9	GaAsP band gaps	$E_\Gamma = 1.424 + 1.150x + 0.176x^2$, $E_X = 1.900 + 0.144x + 0.211x^2$ eV; direct below $x = 0.45$, indirect above. Without N: efficient emission only below 0.45. With the N isoelectronic trap: electron localised in space, spread in k, bound exciton, phonon-free recombination: efficient orange, yellow, green LEDs above 0.45.
15	17-18 3(b)	9	Lattice matching	Critical thickness: largest thickness a mismatched layer grows strained without misfit dislocations, $h_c \approx a_s/2\lvert f\rvert$; above it dislocations lower lifetime, efficiency and mobility. GaAs direct gap: vertical transitions, strong absorption and emission. Si indirect: phonon needed, weak absorption near the edge, very weak emission.
PN junction				
16	20-21 4(c)	5	Graded junction	Diffuse acceptors into a uniformly doped n-type wafer (or grade the doping during epitaxy); near the depth where $N_a = N_d$ the net doping is $a(x - x_j)$: linear. Then $\rho = eax$, the field is parabolic, $W = [12\epsilon_s(V_{bi} + V_R)/(ea)]^{1/3}$ and $C' \propto (V_{bi} + V_R)^{-1/3}$.
Metal-semiconductor contacts				
17	20-21 3(a)	5	Schottky junction	Ideal metal/n-type contact: the barrier from the metal stays $e\phi_{B0} = e(\phi_m - \chi)$. Reverse bias raises the barrier seen from the semiconductor to $e(V_{bi} + V_R)$, widens the depletion layer, tiny current. Forward bias lowers it to $e(V_{bi} - V_a)$, narrows the layer, electrons flow into the metal.

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No.	Paper	Marks	Subtopic	Points to cover
18	20-21 4(b)	8	Barrier lowering	(i) The image charge of an electron lowers the barrier by $\Delta\phi = \sqrt{eE/4\pi\epsilon_s}$ at $x_m = \sqrt{e/16\pi\epsilon_s E}$; E grows with reverse bias, so the reverse current rises with V_R. (ii) Interface states with a neutral level $e\phi_0$ pin the surface Fermi level, so the barrier depends only weakly on the metal work function (Bardeen limit).
19	20-21 5(d)	4	Ohmic contacts	With $\phi_m < \phi_s$ on n-type, electrons flow from the metal into the semiconductor and form an accumulation layer; the bands bend down by $e(\phi_s - \phi_m)$; electrons meet no barrier either way, so the I-V is linear with low resistance: an ohmic contact.
20	18-19 3(a)	4	Ohmic contacts	Specific contact resistance $R_c = (\partial J/\partial V)^{-1}_{V=0}$ (Ω cm^2). Thermionic: $R_c = (kT/e)e^{e\phi_{Bn}/kT}/(A^*T^2)$, exponential in the barrier. Tunnelling ($N_d \gtrsim 10^{19}$): $R_c \propto \exp\left[2\sqrt{\epsilon_s m^*}\phi_{Bn}/(\hbar\sqrt{N_d})\right]$. Also temperature, m^*, ϵ_s, interface quality.
21	18-19 3(c)	3	Ohmic contacts	Barrier-free contact: $\phi_m < \phi_s$ on n-type or $\phi_m > \phi_s$ on p-type, majority carriers accumulate. Tunnelling contact: heavy surface doping makes the barrier a few nm thick, carriers tunnel through.
22	18-19 TT1 7(a)	4	Ohmic contacts	Ohmic: linear symmetric I-V, small R_c. Types: barrier-free ($\phi_m < \phi_s$ on n-type, $\phi_m > \phi_s$ on p-type, accumulation) and tunnelling (heavy surface doping, thin barrier, R_c falls with $\sqrt{N_d}$).
23	17-18 3(c)	7	Ohmic contacts	Metal on n$^+$ ($\gtrsim 10^{19}$ cm^{-3}): depletion width $\propto N_d^{-1/2}$ falls to a few nm; electrons tunnel (field emission) through the barrier in both directions; linear I–V, small R_c: ohmic.
Heterojunctions				
24	18-19 3(b)	6	Resonant tunnelling	Double barrier with a quantum well: levels E_1, E_2. Bias lowers the levels; current peaks when E_1 lies in the band of emitter electrons (resonant tunnelling) and falls when it drops below the emitter E_c (NDR); E_2 gives a second peak and a second NDR region.
25	18-19 6(a)	4	HEMT	2DEG in undoped GaAs separated from its donors (modulation doping): high mobility (8500 cm^2/V s at 300 K, far more at 77 K), high speed (f_T above 100 GHz), high transconductance (about 250 mS/mm), lowest noise, high sheet density, low-voltage operation.
26	18-19 6(b)	7	HEMT	N-AlGaAs / spacer / undoped GaAs: electrons drop by ΔE_c into a triangular well, a 2DEG; the Schottky gate sets $n_s = \epsilon_N(V_g - V_{off})/[e(d + \Delta d)]$ with $V_{off} = \phi_B - \Delta E_c/e - eN_d d_d^2/2\epsilon_N$; negative V_g empties the well, positive fills it; I_D saturates at v_{sat}.
27	17-18 5(b)	5	HEMT	2DEG: electrons quantised across a narrow well, free in the plane. N-AlGaAs donors give electrons to the triangular well in undoped GaAs (modulation doping, high mobility); gate sets $n_s = \epsilon_N(V_g - V_{off})/[e(d + \Delta d)]$; negative V_g empties the well, positive fills it until the AlGaAs conducts.
Bipolar transistors				
28	20-21 6(b)	8	High-level injection	High-level injection: injected electrons in the base approach N_B, extra holes come in for neutrality, effective base doping rises, injection efficiency falls, $I_C \propto e^{V_{BE}/2V_t}$, β falls at high current. Current crowding: the lateral base current's IR drop makes V_{BE} largest at the emitter edges, so current crowds there (hot spots). Cure: narrow interdigitated emitter fingers, more base contacts, ballast resistors.
29	18-19 5(b)	6	Non-ideal effects	Early effect: B-C depletion widens with V_{CB}, neutral base narrows, I_C rises, curves extrapolate to $-V_A$, $r_o = V_A/I_C$; punch-through. Current crowding: lateral base IR drop makes V_{BE} largest at the emitter edges; current crowds there; cure: interdigitated narrow emitters, ballast.

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No.	Paper	Marks	Subtopic	Points to cover
30	18-19 5(c)	7	Gummel-Poon	Charge-control model: $J_n = -eD_n n_i^2 e^{eV_{BE}/kT} / \int_0^{x_B} p\,dx$ (base Gummel number); $I_C = (I_S/q_b)(e^{V_{BE}/V_t} - e^{V_{BC}/V_t})$, $q_b = (q_1/2)(1 + \sqrt{1+4q_2})$: q_1 Early effect, q_2 high injection; ideal and non-ideal base diodes; r_B, r_E, r_C.
31	17-18 4(a)	8	Ebers-Moll	Two diodes $I_F = I_{ES}(e^{V_{BE}/V_t} - 1)$, $I_R = I_{CS}(e^{V_{BC}/V_t} - 1)$ and sources $\alpha_F I_F$, $\alpha_R I_R$; $I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$. Forward active: $I_C = \beta_F I_B$; saturation: both diodes on; cut-off; inverse.
32	17-18 4(b)	3	Bandgap narrowing	At high doping the impurity levels form a band that merges with E_c; many-body effects lower the edges; $E_g = E_{g0} - \Delta E_g$ (about 80 meV at 10^{19} cm^{-3} in Si); p_{E0} rises by $e^{\Delta E_g/kT}$, so γ and β fall.
33	17-18 5(a)	4	TEBT	Thin (2–5 nm) AlGaAs tunnel barrier between n$^+$ emitter and p$^+$ base. Electrons (small m^*, small ΔE_c) tunnel through; holes (heavy, larger ΔE_v) are reflected. $\gamma \to 1$ with a heavily doped base: high gain, low base resistance, low turn-on voltage.
34	17-18 5(c)	11	HBT	Kirk effect: $n_c = J_C/(ev_{sat})$ cancels N_C; for $n_c > N_C$ the field peak moves to the n$^-$/n$^+$ interface and the base widens to $W_B + W_C$; β, f_T fall; $J_K \approx ev_{sat}N_C$. HBT: heavy base doping lets N_C rise (DHBT). Abrupt: spike, higher turn-on, gain $\exp(\Delta E_v/kT)$, hot-electron launch; graded: no spike, full $\exp(\Delta E_g/kT)$, lower turn-on, higher β.
JFETs				
35	20-21 5(a)	6	Channel profiles	The gate–channel reverse bias grows from source to drain, so the depletion layer is a wedge, thickest at the drain. (i) $V_d < V_{d,sat}$: channel open throughout, linear region. (ii) $V_d = V_{d,sat}$: the depletion just touches the bottom at the drain (pinch-off). (iii) $V_d > V_{d,sat}$: the pinched-off part grows towards the source, current saturates.
36	17-18 6(d)	6	Transconductance	$g_m = \partial I_D/\partial V_{GS}$ at constant V_{DS}; saturation $g_{ms} = -(2I_{DSS}/V_p)(1 - V_{GS}/V_p)$. Small-signal circuit: $C_{gs} \parallel r_{gs}$, $C_{gd} \parallel r_{gd}$, $g_m v'_{gs}$ with r_{ds}, C_{ds}, series r_s, r_d; with r_s: $g'_m = g_m/(1 + g_m r_s)$.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Band structure and materials				
1	20-21 1(c)	6	Hole bands	Parabolic band $E = E_v - \hbar^2k^2/2m^*$; slope $\hbar^2k/m^*$; ratio of slopes at the same k is m_{hh}/m_{lh}; Luttinger masses along [100] (γ_2) and [111] (γ_3); ratios 1.45 and 5.30, both above 1.
2	20-21 TT1 7(c)	4	Hole bands	Parabolic band $E = E_v - \hbar^2k^2/2m^*$; $\lvert dE/dk\rvert = \hbar^2k/m^*$; ratio at the same k is m_{hh}/m_{lh}; Luttinger masses along [100] (γ_2) and [111] (γ_3); ratios 1.45 and 5.30, both above 1.
3	17-18 1(b)	4	Effective mass	Band edge as a parabola (Taylor expansion); compare with $E = \hbar^2k^2/2m$ to get $1/m^* = (1/\hbar^2)d^2E/dk^2$; check with $a = (1/\hbar)(d^2E/dk^2)(dk/dt)$ and $\hbar\,dk/dt = F$, so $F = m^*a$; method: fit a parabola (finite difference) to E–k points at the extremum.
Metal-semiconductor contacts				
4	21-22 TT1 7(c)	9	Schottky junction	Assumptions (ideal contact, uniform N_d, depletion approximation, metal a perfect conductor); $\rho = +eN_d$ for $0 < x < x_n$, sheet $-eN_dx_n$ on the metal; Poisson $dE/dx = eN_d/\epsilon_s$, integrate, $E(x_n) = 0$: $E = -(eN_d/\epsilon_s)(x_n - x)$; $E = 0$ in the neutral region and in the metal (Gauss check at $x = 0$); $\lvert E_{max}\rvert = eN_dx_n/\epsilon_s$; $V_{bi} = eN_dx_n^2/2\epsilon_s$ gives x_n.

No.	Paper	Marks	Derivation	Outline
Heterojunctions				
5	20-21 1(d)	8	Band diagrams	E_g of AlGaAs from x, $\Delta E_c = \Delta E_v = 0.156$ eV; band diagrams before (common vacuum level) and after contact (flat E_F, spike and step); hole concentration on each side; solve for E_F; $eV_{bi} = E_{FN} - E_{FP}$; put $E_{vN} - E_{vP} = -\Delta E_v$.
6	20-21 4(a)	12	Fields and potentials	$\rho = +eN_{d1}$, $-eN_{a2}$; Poisson in each material with its own ϵ; fields zero at the edges; $\epsilon_1 E_1(0) = \epsilon_2 E_2(0)$ gives $N_{d1}x_n = N_{a2}x_p$; integrate again with ϕ continuous; $V_{bi} = V_{bi1} + V_{bi2}$; solve for x_n, x_p.
7	20-21 TT1 7(d)	10	Band diagrams	E_g of AlGaAs from x, $\Delta E_c = \Delta E_v = 0.156$ eV; band diagrams before (common vacuum level) and after contact (flat E_F, spike and step); hole concentration on each side; solve for E_F; $eV_{bi} = E_{FN} - E_{FP}$; put $E_{vN} - E_{vP} = -\Delta E_v$.
8	18-19 2(b)	7	Fields and potentials	Poisson on each side with its own ϵ; $E = 0$ at $-x_n$ and x_P; $\epsilon_n E_n(0) = \epsilon_P E_P(0)$ gives $N_{dn}x_n = N_{aP}x_P$; $V_{bin} = eN_{dn}x_n^2/2\epsilon_n$, $V_{biP} = eN_{aP}x_P^2/2\epsilon_P$; add; eliminate one width; x_n, x_P, W.
MOS capacitor				
9	18-19 4(d)	6	Threshold voltage	Flat band: $Q'_m = -Q'_{ss}$, $\phi_s = 0$, so $V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$; threshold: $Q'_{mT} = \lvert Q'_{SD}(\max)\rvert - Q'_{ss}$, $\phi_s = 2\phi_{fp}$; $V_T = V_{FB} + \lvert Q'_{SD}(\max)\rvert/C_{ox} + 2\phi_{fp}$.
Bipolar transistors				
10	20-21 6(a)	9	Ebers-Moll	Two diodes I_F, I_R with sources $\alpha_F I_F$, $\alpha_R I_R$; KCL gives I_C, I_E, I_B; saturation: drop the -1 terms, V_{BE} and V_{BC} as logarithms; solve for I_F, I_R from I_C, I_B; $V_{CE} = V_{BE} - V_{BC}$; reciprocity $I_{CS}/I_{ES} = \alpha_F/\alpha_R$.
11	18-19 2(c)	8	Base minority carriers	Diffusion equation in the base; cosh/sinh general solution; law of the junction: $\delta n_B(0) = n_{B0}(e^{eV_{BE}/kT} - 1)$, $\delta n_B(x_B) = -n_{B0}$; solve for the constants; sinh identity; narrow base gives a straight line.
12	18-19 4(b)	7	HBT	Short-base $J_{nE} = eD_{nB}n_{iB}^2 e^{eV_{BE}/kT}/(N_{AB}W_B)$; short-emitter $J_{pE} = eD_{pE}n_{iE}^2 e^{eV_{BE}/kT}/(N_{DE}W_E)$; ratio; $n_{iB}^2/n_{iE}^2 = e^{\Delta E_g/kT}$.
13	18-19 5(a)	7	Injection efficiency	Diffusion equation in the emitter with $\delta p_E(x_E) = 0$; J_{pE} with $1/\tanh(x_E/L_E)$; base solution gives J_{nE} with $1/\tanh(x_B/L_B)$; $\gamma = J_{nE}/(J_{nE} + J_{pE})$; drop small terms; short-region form.
14	17-18 4(c)	9	Breakdown	$I_{CEO} = M(\alpha I_{CEO} + I_{CBO})$; $I_{CEO} = MI_{CBO}/(1 - \alpha M)$; breakdown at $\alpha M = 1$; $M = 1/[1 - (V/BV_{CBO})^n]$; $BV_{CEO} = BV_{CBO}(1 - \alpha)^{1/n} = BV_{CBO}/\sqrt[n]{\beta}$.
JFETs				
15	20-21 5(b)	8	Pinch-off	Poisson with $\rho = +eN_d$ in the depletion layer; $E = 0$ at $y = h$; potential $eN_d h^2/2\epsilon_s$; equals $V_{bi} - V_{gs}$; gives h; pinch-off at $h = a$; $V_{p0} = ea^2N_d/2\epsilon_s$, $V_p = V_{bi} - V_{p0}$.
16	18-19 6(d)	4	Cut-off frequency	$f_T = g_m/2\pi C_G$; $g_m(\max) = e\mu_n N_d W a/L$; $C_G(\min) = \epsilon_s WL/a$; $f_T(\max) = e\mu_n N_d a^2/(2\pi\epsilon_s L^2) \propto N_d$.
17	17-18 6(b)	6	Pinch-off	Poisson with $\rho = +eN_d$; $E(h) = 0$ gives $E = -eN_d(h - y)/\epsilon_s$; $\phi(h) = eN_d h^2/2\epsilon_s = V_{bi} - V_{gs}$; pinch-off at $h = a$: $V_{p0} = ea^2N_d/2\epsilon_s$.

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7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Alloys and strain					
1	20-21 2(d)	7	Quaternary alloys	Read the corner binaries from Fig. 1; Vegard's law (bilinear interpolation) for $a(x,y)$ and $E_g(x,y)$ of $In_xGa_{1-x}As_ySb_{1-y}$; set $a = 5.93$ Å and $E_g = 1$ eV; solve the two equations.	$x \approx 0.24$, $y \approx 0.60$ (straight-line interpolation; with the bowing printed in Fig. 1 no alloy has both values)
2	17-18 1(d)	9	Quaternary alloys	$E_g = hc/\lambda$; Vegard bilinear $a(x,y)$ and $E_g(x,y)$ from the table; lattice condition $a = 5.93$ Å gives $y(x)$; E_g along it; closest point; f on InP, $h_c \approx a_s/(2f)$.	$E_g = 1.0$ eV not reachable on $a = 5.93$ Å; closest $x = 0$, $y = 0.36$ ($E_g = 0.965$ eV); $f = 1.04$ %, $h_c \approx 28$ nm
3	17-18 TT1 7(a)	8	Lattice matching	$h_c = a_s/(2f)$ gives f; $a_L = a_s(1+f)$; Vegard $x = (a_L - a_{GaAs})/(a_{InAs} - a_{GaAs})$.	$f = 5.65$ %, $a_L = 5.973$ Å, $x = 0.79$
Metal-semiconductor contacts					
4	20-21 3(c)	8	Schottky junction	$\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$; $C' = \epsilon_s/x_n$.	$\phi_{B0} = 0.54$ V; $V_{bi} = 0.334$ V; $\lvert E_{max}\rvert = 3.22 \times 10^4$ V/cm; $C' = 4.98 \times 10^{-8}$ F/cm^2
5	20-21 3(d)	6	Barrier lowering	Image-force lowering $\Delta\phi = \sqrt{eE/(4\pi\epsilon_s)}$; position $x_m = \sqrt{e/(16\pi\epsilon_s E)}$ with ϵ_s of GaAs.	$\Delta\phi = 27.3$ mV at $x_m = 2.01$ nm
6	18-19 1(c)	6	Schottky junction	$\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$.	$\phi_{B0} = 0.97$ V; $V_{bi} = 0.790$ V; $\lvert E_{max}\rvert = 4.23 \times 10^4$ V/cm
7	18-19 3(d)	0	Schottky junction	n-type: $\phi_{Bn} = \phi_m - \chi$; p-type: $\phi_{Bp} = E_g/e + \chi - \phi_m$; positive barrier means Schottky, zero or negative means ohmic.	Al/n-GaAs ohmic (-0.01 V); Cu/n-GaAs Schottky 0.91 V; Ag/p-GaAs Schottky 0.849 V; Au/p-GaAs Schottky 0.119 V (nearly ohmic)
8	18-19 TT1 7(b)	6	Schottky junction	$\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$.	$\phi_{B0} = 0.97$ V; $V_{bi} = 0.790$ V; $\lvert E_{max}\rvert = 4.23 \times 10^4$ V/cm
9	17-18 2(c)	10	Schottky junction	$\phi_{B0} = \phi_m - \chi$; $\phi_n = V_t \ln(N_c/N_d)$; $V_{bi} = \phi_{B0} - \phi_n$; $x_n = [2\epsilon_s V_{bi}/(eN_d)]^{1/2}$; $\lvert E_{max}\rvert = eN_d x_n/\epsilon_s$.	$\phi_{B0} = 0.48$ V; $V_{bi} = 0.362$ V; $x_n = 0.324$ μm; $\lvert E_{max}\rvert = 2.24 \times 10^4$ V/cm

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No.	Paper	Marks	Subtopic	Method	Answer
Heterojunctions					
10	18-19 2(d)	5	Fields and potentials	$\Delta E_c = \chi_n - \chi_P$, $\Delta E_v = \Delta E_g - \Delta E_c$; solve $eV_{bi} = \Delta E_v + kT\ln(N_{aP}N_{vn}/(p_{n0}N_{vP}))$ for p_{n0}; $N_d = n_i^2/p_{n0}$; x_P from the heterojunction formula.	$\Delta E_c = 0.06$ eV, $\Delta E_v = 0.70$ eV; $N_d = 7.2\times10^{15}$ cm^{-3}; $x_P = 0.261$ μm
11	17-18 2(a)	7	Band diagrams	Anderson: $\Delta E_c = \chi_n - \chi_P$, $\Delta E_v = \Delta E_g - \Delta E_c$; $p_{n0} = n_i^2/N_d$; $eV_{bi} = \Delta E_v + kT\ln(p_{P0}N_{vn}/(p_{n0}N_{vP}))$.	$\Delta E_c = 0.06$ eV, $\Delta E_v = 0.70$ eV, $V_{bi} = 1.01$ V
12	17-18 5(d)	5	HEMT	$V_{p2} = eN_d d_d^2/(2\epsilon_N)$; $V_{off} = \phi_B - \Delta E_c/e - V_{p2}$; $n_s = \epsilon_N(V_g - V_{off})/[e(d_d + d_i + \Delta d)]$ with $\Delta d = 80$ Å.	$V_{p2} = 1.85$ V; $V_{off} = -1.22$ V; $n_s = 1.37\times10^{12}$ cm^{-2}
MOS capacitor					
13	18-19 TT2 8(b)	5	Threshold voltage	$\phi_{fp} = V_t\ln(N_a/n_i)$; $x_{dT} = (4\epsilon_s\phi_{fp}/eN_a)^{1/2}$; $\lvert Q'_{SD}(\max)\rvert = eN_a x_{dT}$; $V_{TN} = (\lvert Q'_{SD}(\max)\rvert - eQ'_{ss})t_{ox}/\epsilon_{ox} + \phi_{ms} + 2\phi_{fp}$.	$\phi_{fp} = 0.228$ V, $x_{dT} = 2.43$ μm, $V_{TN} = 0.546$ V
Bipolar transistors					
14	18-19 5(d)	5	Injection efficiency	$L = \sqrt{D\tau}$; $\alpha = \beta/(1+\beta)$; take the minimum x_B, $\alpha_T = 1/\cosh(x_B/L_B)$; $\gamma = \alpha/(\alpha_T\delta)$; solve the γ formula for N_E.	$x_B = 0.80$ μm, $N_E = 4.6\times10^{18}$ cm^{-3}; $\alpha_T = 0.99930$, $\gamma = 0.99656$
15	17-18 4(d)	5	Bandgap narrowing	$p_{E0} = n_i^2/N_E$; read ΔE_g from Fig. 1; $p'_{E0} = p_{E0}\exp(\Delta E_g/kT)$.	$\Delta E_g \approx 23$ and 78 meV; $p'_{E0} = 549$ and 454 cm^{-3}; ratio 2.4 and 20
JFETs					
16	20-21 5(c)	7	Pinch-off	$V_{bi} = V_t\ln\left(N_aN_d/n_i^2\right)$; $V_{p0} = ea^2N_d/2\epsilon_s$; $V_p = V_{bi} - V_{p0}$; for an open channel $a - h = 0.10$ μm, $h = 0.60$ μm, $V_{bi} - V_{gs} = eN_dh^2/2\epsilon_s$, solve for V_{gs}.	$V_{bi} = 1.250$ V; $V_{p0} = 1.016$ V; $V_p = +0.235$ V; $V_{gs} \approx +0.50$ V (forward)
17	20-21 6(c)	8	Transconductance	$I_{P1} = \mu_n(eN_d)^2Wa^3/(6\epsilon_sL)$; V_{bi} and V_{p0}; $I_D(sat)$ at $V_{GS} = 0$ from the JFET saturation equation; $g_{ms} = (3I_{P1}/V_{p0})(1 - \sqrt{(V_{bi} - V_{GS})/V_{p0}})$ at $V_{GS} = 0$.	$I_{P1} = 0.522$ mA ($I_{D1}(sat) = 0.3135$ mA at $V_{GS} = 0$); $g_{ms}(max) = 0.204$ mS
18	18-19 6(e)	5	Pinch-off	$\phi_n = V_t\ln(N_c/N_d)$; $V_{bi} = \phi_{Bn} - \phi_n$; $V_{p0} = V_{bi} - V_T$; $a = (2\epsilon_sV_{p0}/eN_d)^{1/2}$.	$V_{bi} = 0.749$ V, $V_{p0} = 0.649$ V, $a = 0.686$ μm
19	17-18 6(c)	5	Pinch-off	$V_{p0} = ea^2N_a/(2\epsilon_s)$; $V_{bi} = V_t\ln\left(N_aN_d/n_i^2\right)$; p-channel: $V_p = V_{p0} - V_{bi}$.	$V_{p0} = 1.236$ V; $V_{bi} = 0.814$ V; $V_p = 0.422$ V

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8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Band structure and materials			
1	21-22 TT1 7(b)	Diagram	bulk, quantum well, wire and dot
2	21-22 TT1 7(b)	Graph	density of states against energy in 3D, 2D, 1D and 0D
3	20-21 1(c)	Graph	E-k of the Si heavy- and light-hole bands along [111] and [100]
4	20-21 2(a)	Graph	fibre loss windows and the wavelength ranges of the materials
5	20-21 3(b)	Diagram	counting allowed k points in 1D, 2D and 3D
6	20-21 3(b)	Graph	density of states against energy in 3D, 2D, 1D and 0D
7	20-21 TT1 7(c)	Graph	E-k of the Si heavy- and light-hole bands along [111] and [100]
8	18-19 1(a)	Diagram	hexagonal close-packed cell
9	18-19 1(a)	Diagram	zinc blende (GaAs) cell with tetrahedral bonds
10	18-19 1(a)	Diagram	body-centred cubic cell
11	17-18 1(a)	Diagram	atoms of a single-crystal, a polycrystalline and an amorphous solid
12	17-18 1(b)	Graph	two E–k parabolas of different curvature
13	17-18 1(b)	Graph	parabola fitted to E–k points
High-field transport			
14	20-21 1(a)	Diagram	electron drift with and without a field
15	20-21 1(a)	Diagram	simplified E-k diagrams of Si and GaAs
16	20-21 1(b)	Graph	drift velocity against field for Si and GaAs
17	20-21 1(b)	Diagram	the two GaAs valleys and electron transfer
18	20-21 1(b)	Diagram	a travelling domain in a Gunn diode and the current pulses
19	20-21 TT1 7(a)	Diagram	simplified E-k diagrams of Si and GaAs
20	20-21 TT1 7(b)	Diagram	the two GaAs valleys and electron transfer
21	20-21 TT1 7(b)	Diagram	a travelling domain in a Gunn diode and the current pulses
22	17-18 TT1 7(b)	Graph	E–k of Si and GaAs
23	17-18 TT1 7(b)	Graph	drift velocity against field for Si and GaAs
Alloys and strain			
24	21-22 TT1 7(a)	Graph	direct and indirect gaps of GaAsP against x
25	21-22 TT1 7(a)	Diagram	simplified E-k of GaAs, the crossover and GaP
26	20-21 2(b)	Diagram	matched, strained and relaxed layers on a substrate
27	20-21 2(b)	Graph	critical thickness against misfit
28	20-21 2(c)	Graph	GaAsP band gap against x with direct and indirect branches
29	20-21 2(d)	Graph	InGaAsSb region of Fig. 1 with the target point
30	18-19 1(d)	Diagram	horizontal VPE reactor with its temperature profile
31	17-18 1(c)	Graph	E_Γ and E_X of $GaAs_{1-x}P_x$ against x
32	17-18 1(d)	Graph	lattice and band-gap lines in the (x, y) square
33	17-18 3(b)	Diagram	matched, strained and relaxed layers
34	17-18 3(b)	Diagram	E–k of Si and GaAs with the transitions

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No.	Paper	Kind	Diagram
PN junction			
35	20-21 4(c)	Diagram	graded junction by diffusion
36	20-21 4(c)	Graph	diffused acceptor profile and net doping
37	20-21 4(c)	Graph	linear charge, parabolic field
Metal-semiconductor contacts			
38	21-22 TT1 7(c)	Diagram	metal and n-semiconductor before contact and in equilibrium
39	21-22 TT1 7(c)	Graph	charge, field and potential across the junction
40	20-21 3(a)	Diagram	metal–n-semiconductor band diagrams at zero, reverse and forward bias
41	20-21 3(c)	Diagram	W and n-Si before and after contact
42	20-21 3(c)	Graph	space charge and field in the depletion region
43	20-21 3(d)	Diagram	electron and its image charge, and the lowered barrier
44	20-21 4(b)	Graph	barrier with image-force lowering
45	20-21 4(b)	Diagram	metal–semiconductor with an interfacial layer and interface states
46	20-21 4(b)	Graph	barrier height against metal work function
47	20-21 5(d)	Diagram	metal and n-semiconductor band diagrams for $\phi_m < \phi_s$
48	20-21 5(d)	Graph	ohmic and rectifying I-V curves
49	18-19 1(b)	Diagram	metal–semiconductor band diagram with an interfacial layer and interface states
50	18-19 1(c)	Diagram	Au / n-Ge band diagram with the values
51	18-19 1(c)	Graph	electric field in the depletion region
52	18-19 3(a)	Graph	specific contact resistance against doping for several barriers
53	18-19 3(c)	Diagram	ohmic contacts by accumulation and by tunnelling
54	18-19 3(d)	Diagram	band diagrams of the four metal–GaAs junctions
55	18-19 TT1 7(a)	Diagram	ohmic contacts by accumulation and by tunnelling
56	18-19 TT1 7(b)	Diagram	Au / n-Ge band diagram with the values
57	17-18 2(c)	Diagram	W / n-GaAs band diagram with the values
58	17-18 3(a)	Diagram	metal and n-type semiconductor before and after contact ($\phi_m > \phi_s$)
59	17-18 3(a)	Diagram	metal and p-type semiconductor before and after contact ($\phi_m < \phi_s$)
60	17-18 3(c)	Diagram	tunnelling junction band diagram
Heterojunctions			
61	20-21 1(d)	Diagram	the three heterojunction band alignments
62	20-21 1(d)	Diagram	p-GaAs and N-AlGaAs before contact
63	20-21 1(d)	Diagram	the heterojunction after contact in equilibrium
64	20-21 4(a)	Diagram	narrow-gap N and wide-gap P before contact
65	20-21 4(a)	Diagram	equilibrium band diagram of the straddling nP junction
66	20-21 4(a)	Graph	charge, field and potential with the field jump at $x = 0$
67	20-21 TT1 7(d)	Diagram	p-GaAs and N-AlGaAs before contact
68	20-21 TT1 7(d)	Diagram	the heterojunction after contact in equilibrium
69	18-19 2(a)	Diagram	pN (p-GaAs/N-AlGaAs) equilibrium band diagram
70	18-19 2(a)	Diagram	nN (n-GaAs/N-AlGaAs) band diagram with a notch
71	18-19 2(b)	Graph	charge, field and potential of a heterojunction
72	18-19 2(d)	Diagram	n-Ge / P-GaAs band diagram with the values
73	18-19 3(b)	Diagram	RTD band diagrams at four biases
74	18-19 3(b)	Graph	I-V with two peaks
75	18-19 6(b)	Diagram	HEMT layer structure

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No.	Paper	Kind	Diagram
76	18-19 6(b)	Diagram	conduction band under the gate with the 2DEG
77	17-18 2(a)	Diagram	n-Ge / p-GaAs equilibrium band diagram
78	17-18 2(b)	Diagram	N^+-AlGaAs / intrinsic GaAs band diagram with a 2DEG notch
79	17-18 2(b)	Diagram	P^+-AlGaAs / n-GaAs band diagram
80	17-18 5(b)	Diagram	HEMT layer structure
81	17-18 5(b)	Diagram	conduction band under the gate with the 2DEG well
MOS capacitor			
82	18-19 4(c)	Graph	charge distribution in an MOS capacitor at inversion
83	18-19 4(c)	Graph	electric field and potential across oxide and silicon
84	18-19 4(c)	Graph	high- and low-frequency C-V curves
85	18-19 4(d)	Diagram	MOS band diagrams at flat band and at threshold
86	18-19 TT2 8(a)	Diagram	MOS band diagrams in accumulation, depletion and inversion for p- and n-type substrates
Bipolar transistors			
87	20-21 6(a)	Diagram	npn regions and currents
88	20-21 6(a)	Circuit	Ebers-Moll equivalent circuit
89	20-21 6(a)	Graph	output curves with the saturation region, and $V_{CE}(sat)$ against I_B
90	20-21 6(b)	Graph	Gummel plot with high-level injection
91	20-21 6(b)	Diagram	emitter current crowding under the emitter
92	20-21 6(b)	Diagram	interdigitated emitter-base layout
93	18-19 2(c)	Graph	minority carriers in E, B, C in forward-active mode
94	18-19 2(c)	Graph	exact base profile for several x_B/L_B
95	18-19 4(a)	Diagram	DHBT mesa layer structure
96	18-19 4(a)	Graph	DHBT doping profile and Al fraction
97	18-19 4(b)	Diagram	HBT band diagram in forward-active mode
98	18-19 5(a)	Diagram	current components in an npn transistor
99	18-19 5(b)	Graph	base-width modulation and output curves with the Early voltage
100	18-19 5(b)	Diagram	emitter current crowding
101	18-19 5(c)	Circuit	Gummel-Poon equivalent circuit
102	18-19 5(c)	Graph	Gummel plot and gain against current
103	18-19 6(c)	Circuit	Ebers-Moll equivalent circuit
104	17-18 4(a)	Circuit	Ebers–Moll equivalent circuit
105	17-18 4(a)	Diagram	npn modes of operation
106	17-18 4(c)	Graph	I_C against V_{CE} showing $BV_{CEO} < BV_{CBO}$
107	17-18 4(d)	Graph	bandgap narrowing against donor doping (Fig. 1)
108	17-18 5(a)	Diagram	TEBT band diagram in forward-active mode
109	17-18 5(c)	Graph	charge and field in the collector at low and high current
110	17-18 5(c)	Diagram	bands with base push-out
111	17-18 5(c)	Diagram	abrupt and graded HBT emitter-base junctions
JFETs			
112	20-21 5(a)	Diagram	channel space-charge profiles in the three regions
113	20-21 5(a)	Graph	drain current against drain voltage with the three points
114	20-21 5(b)	Diagram	channel geometry of single- and two-gate JFETs
115	20-21 5(b)	Graph	charge, field and potential across the gate junction
116	20-21 5(c)	Graph	depletion width against gate voltage
117	20-21 6(c)	Graph	transfer curve and transconductance
118	18-19 6(d)	Diagram	MESFET cross-section with a, L, h
119	17-18 6(a)	Diagram	JFET depletion region at $V_d = 0$, below, at and above $V_{d,sat}$
120	17-18 6(b)	Diagram	MESFET cross-section with a and h
121	17-18 6(d)	Circuit	JFET small-signal equivalent circuit

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9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Band structure and materials		
Single crystal vs polycrystalline vs amorphous	long-range order everywhere vs within grains vs none	17-18 1(a)
hcp vs bcc	close packed, 12 neighbours, packing 0.74 vs 8 neighbours, packing 0.68	18-19 1(a)
3D vs 2D vs 1D density of states	$\sqrt{E}$ vs steps vs $1/\sqrt{E}$ spikes	21-22 TT1 7(b), 20-21 3(b)
High-field transport		
Si vs GaAs electrons	indirect gap, six flat X valleys, $m^* = 0.26\,m_0$ vs direct gap, one curved Γ valley, $m^* = 0.067\,m_0$, 6 times the mobility	20-21 1(a), 20-21 TT1 7(a), 17-18 TT1 7(b)
Velocity saturation vs Gunn effect	velocity levels off (Si) vs velocity falls after a peak, negative mobility, oscillation (GaAs)	20-21 1(b)
Alloys and strain		
Direct vs indirect gap	minimum at $k = 0$, vertical transition, bright vs minimum at the zone edge, needs a phonon, dim	21-22 TT1 7(a), 17-18 3(b)
Strained vs relaxed layer	thinner than h_c, no dislocations, distorted cell vs thicker, misfit dislocations, own lattice constant	20-21 2(b)
PN junction		
Abrupt vs linearly graded junction	step doping, $W \propto V^{1/2}$ vs linear doping, $W \propto V^{1/3}$	20-21 4(c)
Metal-semiconductor contacts		
Rectifying on n vs p	$\phi_m > \phi_s$, bands bend up vs $\phi_m < \phi_s$, bands bend down	17-18 3(a)
Schottky limit vs Bardeen limit	barrier follows $\phi_m - \chi$ vs barrier pinned by interface states	20-21 4(b), 18-19 1(b)
Barrier-free vs tunnelling ohmic contact	no barrier vs thin barrier from heavy doping	18-19 TT1 7(a)
Reverse vs forward biased Schottky junction	barrier $V_{bi} + V_R$, wider depletion, small current vs barrier $V_{bi} - V_a$, narrower, large electron current	20-21 3(a)
Ohmic vs rectifying contact	$\phi_m < \phi_s$ (n-type), no barrier, linear I-V vs $\phi_m > \phi_s$, Schottky barrier, diode I-V	20-21 5(d)
Heterojunctions		
N^+/i vs P^+/n AlGaAs/GaAs	accumulation notch in the GaAs vs depletion of the n-GaAs	17-18 2(b)
pN vs nN heterojunction	both sides depleted vs depletion plus accumulation (2DEG notch)	18-19 2(a)
HEMT vs MESFET	2DEG in undoped GaAs vs electrons among their donors	18-19 6(b)
Straddling vs staggered vs broken-gap heterojunction	one gap inside the other vs offsets in the same direction vs the gaps do not overlap	20-21 1(d)
MOS capacitor		
High- vs low-frequency C-V in inversion	stays at C'_{min} vs returns to C_{ox}	18-19 4(c)
Bipolar transistors		
Abrupt vs graded HBT	spike, higher turn-on, hot electrons vs no spike, lower turn-on, higher gain	17-18 5(c)
SHBT vs DHBT	wide-gap emitter only vs wide-gap emitter and collector	18-19 4(a)
BJT vs HBT	gain needs $N_E \gg N_B$ vs wide-gap emitter, heavily doped base	18-19 4(b)

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Pair	Key difference	Asked in
Early effect vs current crowding	base width changes with V_{CB} vs lateral IR drop under the emitter	18-19 5(b)
Ebers-Moll vs Gummel-Poon	constant two-diode model vs charge-based with Early effect and high injection	18-19 5(c)
High-level injection vs current crowding	base charge rises, β falls vs lateral IR drop, current concentrates at emitter edges	20-21 6(b)

10. Repeated questions:

12 questions came back, exactly or with new numbers; 1 of them in three or more papers. Most repeated first; the latest paper first in each row.

No.	Times	Topic	Subtopic	Asked in (marks)
1	3	High-field transport	Mobility	20-21 1(a) (6), 20-21 TT1 7(a) (3), 17-18 TT1 7(b) (12)
2	2	Alloys and strain	GaAsP band gaps	21-22 TT1 7(a) (3), 17-18 1(c) (9)
3	2	Band structure and materials	Density of states	21-22 TT1 7(b) (8), 20-21 3(b) (6)
4	2	JFETs	Channel profiles	20-21 5(a) (6), 17-18 6(a) (8)
5	2	JFETs	Pinch-off	20-21 5(b) (8), 17-18 6(b) (6)
6	2	High-field transport	Gunn effect	20-21 1(b) (5), 20-21 TT1 7(b) (3)
7	2	Band structure and materials	Hole bands	20-21 1(c) (6), 20-21 TT1 7(c) (4)
8	2	Heterojunctions	Band diagrams	20-21 1(d) (8), 20-21 TT1 7(d) (10)
9	2	Bipolar transistors	Ebers-Moll	18-19 6(c) (5), 17-18 4(a) (8)
10	2	Heterojunctions	HEMT	18-19 6(b) (7), 17-18 5(b) (5)
11	2	Metal-semiconductor contacts	Schottky junction	18-19 1(c) (6), 18-19 TT1 7(b) (6)
12	2	Metal-semiconductor contacts	Ohmic contacts	18-19 3(c) (3), 18-19 TT1 7(a) (4)

11. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram
Band structure and materials	4	3	·	1
High-field transport	5	·	·	·
Alloys and strain	6	·	3	·
PN junction	1	·	·	·
Metal-semiconductor contacts	7	1	6	2
Heterojunctions	4	4	3	2
MOS capacitor	·	1	1	2
Bipolar transistors	7	5	2	2
JFETs	2	3	4	1

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12. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Bipolar transistors	108	21 %
2	Metal-semiconductor contacts	94	40 %
3	Heterojunctions	81	55 %
4	Alloys and strain	66	68 %
5	JFETs	63	81 %
6	Band structure and materials	42	89 %
7	High-field transport	29	95 %
8	MOS capacitor	22	99 %
9	PN junction	5	100 %

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